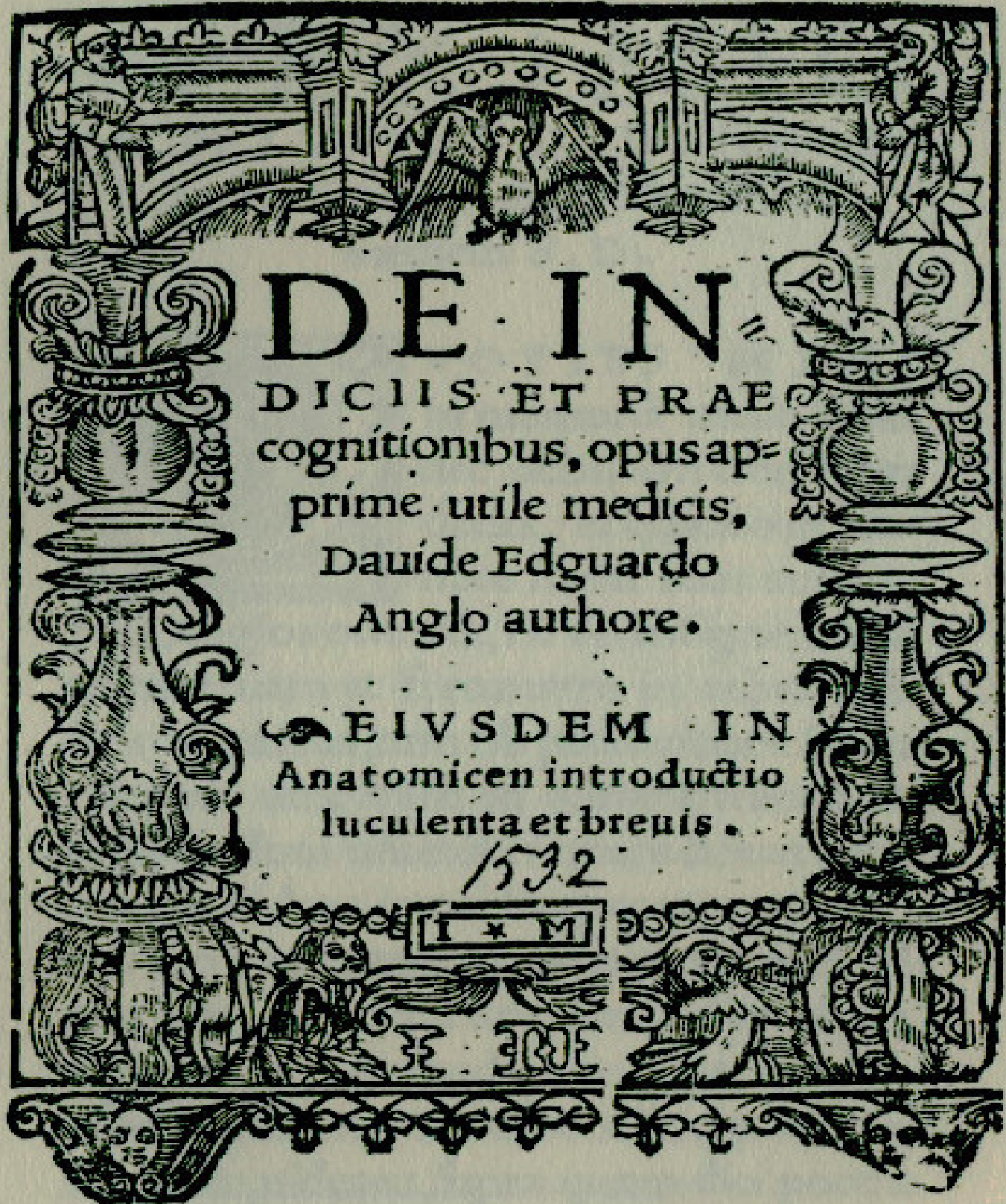
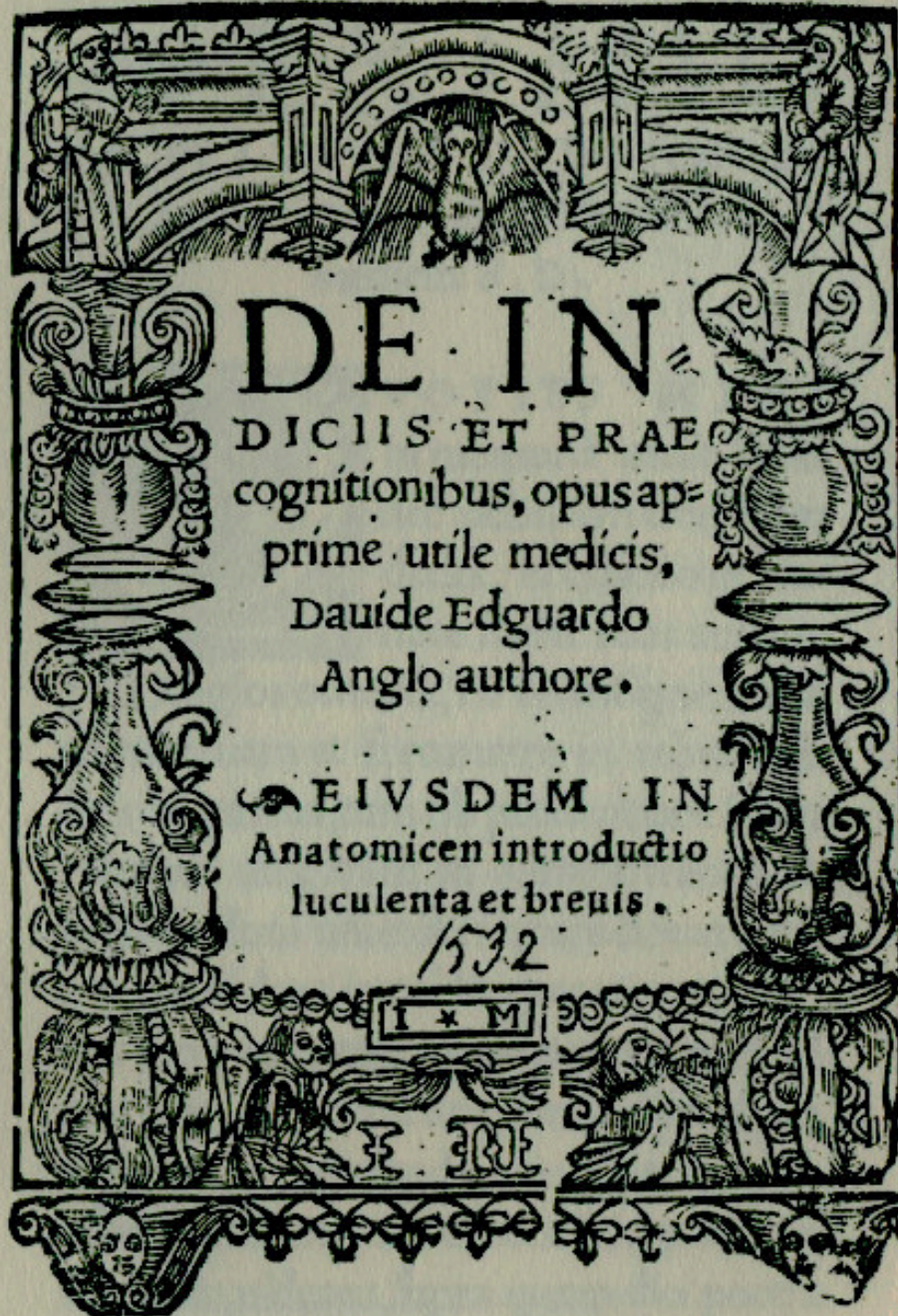


DAVID EDWARDES



INTRODUCTION TO
ANATOMY, 1532

DAVID EDWARDES



INTRODUCTION TO
ANATOMY, 1532

The Project Gutenberg eBook of Introduction to Anatomy, 1532

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Introduction to Anatomy, 1532

Author: David Edguard

Translator: Charles Donald O'Malley
K. F. Russell

Release date: April 25, 2019 [eBook #59357]

Language: English, Latin

Other information and formats: www.gutenberg.org/ebooks/59357

Credits: Produced by Stephen Hutcheson, Dave Morgan and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK
INTRODUCTION TO ANATOMY, 1532 ***

DAVID EDWARDES
Introduction to
Anatomy
1532

A FACSIMILE REPRODUCTION
WITH ENGLISH TRANSLATION AND
AN INTRODUCTORY ESSAY
ON ANATOMICAL STUDIES IN
TUDOR ENGLAND

BY
C. D. O'MALLEY
AND
K. F. RUSSELL

STANFORD UNIVERSITY PRESS
Stanford, California
1961

English translation and Introduction
© C. D. O'Malley and K. F. Russell, 1961

PRINTED IN GREAT BRITAIN
AT THE UNIVERSITY PRESS, OXFORD
BY VIVIAN RIDLER
PRINTER TO THE UNIVERSITY

TO THE MEMORY OF
CHARLES SINGER
FRIEND AND MENTOR

CONTENTS

<u>INTRODUCTION</u>	1
<u>NOTES TO THE INTRODUCTION</u>	25
<u>FACSIMILE REPRODUCTION</u>	31
<u>ENGLISH TRANSLATION</u>	53

Grateful acknowledgements are made for assistance from the National Science Foundation in the preparation of this work; to the British Museum for permission to photograph the only copy of David Edwardes's *Introduction* known to be in existence; and to the Wellcome Trust whose help made the publication of this work possible.

INTRODUCTION

On 22 August 1485 the battle of Bosworth provided its victor with the throne of England. Richard III died sword in hand and was unceremoniously buried in the Grey Friars at Leicester, and on that same day the victor, Henry Tudor, was as simply crowned and acclaimed by his troops as Henry VII. So began the Tudor dynasty in England which was to last until the death of Elizabeth in 1603, to be one of the most colourful periods of English history and to witness the arrival of the Renaissance in England. Later than its manifestation on the Continent, but thereby reaping the benefits of continental developments, English humanism as a result was soon to become no mean rival. The development of English literature is too well known for comment, while classical studies, and especially those in Greek, were to rival their continental counterpart by the end of the first quarter of the sixteenth century. Science, however, and more particularly medicine, were laggards.

In those closing years of the fifteenth century which ushered in the new Tudor monarchy the art of healing derived from two sources, the universities of Oxford and Cambridge and the organizations of barbers and surgeons. At Oxford medical teaching was organized by the fifteenth century, and medicine constituted one of the four faculties of the university together with theology, law, and arts. Yet at Oxford, as at Cambridge, the

medical curriculum was long to remain medieval.^[1] Both schools had taken their model from Paris, but whereas Parisian medicine had begun to stir and advance in the fifteenth century, the English universities remained somnolent. At Cambridge the degree of Doctor of Medicine required altogether twelve years of study based upon lectures and discussions drawn from medieval sources. While it is true that two years of this time were to be spent in the practice of medicine—seemingly a borrowing from the methods of Montpellier—there was no provision for human anatomical study,^[2] although this was recognized and demonstrated

with some slight annual regularity to the Parisian students from the latter fifteenth century onward.

If we turn to the other source of healing, the organizations of the barbers and the surgeons, in so far as anatomy was concerned the situation was no better and, indeed, it may be said to have been worse in view of the obvious relationship which ought to have obtained between surgery and anatomy. In

London the fraternity of barbers existed as early as 1308,^[3] and the craft of surgery as a body distinct from that of the barbers is recorded in 1368.^[4]

Both barbers and surgeons sought to establish rules of professional conduct for the members of their respective organizations as well as a period of time and a curriculum to be satisfied by aspirants to barbering or to surgery. Despite the efforts of the surgeons to control the practice of surgery, relegating to the barbers only the most simple and menial tasks, certain of the more ambitious barbers sought to go beyond such activities as beard-trimming, cutting, and phlebotomy, and this determination gave rise in the first quarter of the fifteenth century to the barber-surgeon[, no longer acting in the normal occupation of the barber and clearly divorced from his old trade.^[5]

Throughout the fifteenth century the barber-surgeons and surgeons appear to have remained on fairly amicable terms, presumably carrying on ^[3] much the same sort of practice. The surgeons, who took precedence on occasions of solemnities and festivities, were perhaps somewhat better trained, but nowhere is there any record that such training required the

study of human anatomy.^[6] One learned the trade by apprenticeship to a surgeon and by consulting textbooks of surgery. From surviving manuscripts it is possible to determine what these textual guides were: primarily such as those of the celebrated fourteenth-century surgeons, Gui de Chauliac and Henri de Mondeville. It is true that late medieval surgeons were accustomed to introduce the surgical subject by a short anatomical preface, medieval in character, the result of cursory and incomplete post-mortem examinations, but hardly sufficient to permit a proper grasp of anatomy even were that possible of attainment from literary sources.

Hence the opening of the Tudor dynasty in England witnessed a medicine and a surgery lacking the essential and fundamental knowledge of the human structure. The traditions of English medicine were medieval, and medieval medicine had not concerned itself especially with anatomy. If we compare continental medicine of the same period the situation is found to be considerably different. In the course of the fifteenth century anatomy was being practised—diffidently to be sure, but nevertheless recognized and employed in Paris where the first human dissection, in the form of a brief autopsy, had been performed in 1407.^[7] The first human anatomy mentioned in the *Commentaries* of the Medical Faculty of Paris was performed in 1477-8 on the body of an executed criminal,^[8] but the incident is recorded without any suggestion of its being a novelty and so raises the possibility that there may have been other dissections in previous years. The practice of human anatomy was even earlier in Italy where there is 4^[9] record of an autopsy in 1286, and in 1316 Mundinus, called the ‘Restorer of Anatomy’, completed his *Anothomia* in which he describes his systematic dissection of the human body. Official decree permitted the practice of human dissection in many cities, especially those with medical schools, and such official recognition was granted at Bologna in 1405^[10] and at Padua in 1429.^[11] Elsewhere similar recognition of human dissection was obtained at Montpellier in 1340,^[12] at Lerida in 1391,^[13] at Vienna in 1435,^[14] and at Tübingen in 1485.^[15] As a consequence, by the opening of the sixteenth century a series of anatomical texts, based in varying degrees upon human dissection, began to appear, such as those of Benedetti, Achillini, and Berengario da Carpi.

The difference can be explained, at least in part, by the fact that on the Continent the classical revival of the Renaissance had caused or was causing medieval tradition to be replaced by that of classical antiquity. The Renaissance represented an effort to revive the spirit and interests of the classical world, and classical antiquity had been much interested in the structure of man. Especially important was the recovery of the Greek language and literature since it made possible the recovery of the writings

of the great classical physicians, notably Hippocrates and Galen, for generally speaking classical Greece had shown more interest in human anatomy than had classical Rome. This recovery had occurred first in Italy, then moved northward across the Alps and only in the early sixteenth century did it reach England.

While even earlier some Englishmen had travelled to Italy to study the classical revival at its source, and even to study the more advanced Italian medicine of Padua, it may be said that Thomas Grocyn was the first significant leader of the new classical movement in England, in particular the recovery of Greek. He had managed to learn some Greek even in England, but it was not until after a trip to Italy in 1488, where he spent two years, that he returned to instil Oxford with an enthusiasm for classical Greek humanism. But if Grocyn is of importance as an English pioneer in the recovery of Greek and Hellenic studies, of far greater importance for the present subject was Grocyn's lifelong friend and ultimately the executor of his estate, Thomas Linacre.

Linacre looms very large in the revival of classical medicine which gave a general impetus toward a better and more modern medicine. Born at Canterbury about 1460 he was led ultimately by his studies to Oxford where he became a fellow of All Souls College in 1484. Although by this time he could make some beginnings of the study of the revived classical literature, and even Greek, at Oxford, nevertheless it was still desirable if possible to pursue such studies in Italy, and with the opportunity offered him, Linacre travelled to that land about 1487, remaining at least until 1496, in which latter year he received the degree of Doctor of Medicine from the University of Padua.

Returned to England, Linacre taught Greek at Oxford. Grocyn was his friend, Sir Thomas More his pupil, and upon the arrival at Oxford of Erasmus, that great classical scholar likewise became an intimate, all of them enthusiasts and promoters of Greek studies.

However, as a physician Linacre had a special bent toward the Greek medical classics. This was manifested by the appearance in 1517 of his translation of Galen's book *On Hygiene*. In 1519 this was followed by the *Method of Treatment*, in 1521 by the book *On Temperaments*, and two years later by the *Natural Faculties* and *On the Use of Pulses*. In 1524 just [6] after Linacre's death a sixth translation, that of Galen's *Differences of Symptoms* and *Causes of Symptoms*, appeared. As yet very few physicians in England knew Greek, but they all knew Latin, and these accurate translations into clear, straightforward Latin made a considerable portion of Galen's medical writings available for the first time. The contrast between medieval medical writings and those of Galen which had now been made available seemed to emphasize that general Renaissance belief that civilization had reached its peak in classical times and that much could be gained by a return to classical teachings, in this instance the teaching of classical physicians. It is true that only the Galenic books on medicine had been translated, but they were sufficient to whet the appetite for more, and as the new generation of physicians arose, now trained in Greek, if the pattern were followed, they would turn to the Galenic writings on anatomy [20] in the original language as well as to those of Hippocrates. The first of this younger generation who is recorded to have come under this Greek medical influence and made this possibility a reality produced two remarkable pioneer efforts: the first recorded dissection of a human body in England about 1531 and the first book on anatomy written in England, published in 1532, or, reckoned according to the Gregorian calendar, 1533.

The person responsible for these two milestones was named David Edwardes, or, in the Latin form he employed, Edguardus. However, very little is known of his life and activities. He was admitted as a scholar to Corpus Christi College, Oxford, on 9 August 1517, and the register of admissions indicates that he was then fifteen years old and a native of [21] Northamptonshire. [22] He became Bachelor of Arts in 1522 although for a time previous to this, in 1521, he appears briefly to have held the [7] readership in Greek, substituting for the regular reader, Edward [23] Wotton, then abroad. [24] In 1525 Edwardes became Master of Arts, and thereafter received a fellowship in the college. He is further mentioned in

the account book of the college for 1527-8 as receiving 38s. 9d.,^[25] presumably for further teaching of Greek.

Corpus Christi College had been founded in 1515-16 chiefly through the magnanimity of Richard Foxe, Bishop of Winchester, and was provided with its statutes in 1517. The founder, strongly interested in the newly revived classical learning had provided for a chair of Greek, which, as has been mentioned, was briefly held by Edwardes in an interim capacity, while the first president of the college, John Claymond, was likewise a strong advocate of the new learning.

Perhaps not sufficient stress has been placed upon the contribution made by physicians, at least in England, to the revival of Greek studies, although it is sometimes difficult to determine which of the two disciplines, medicine or Greek, was the impulsion to the study of the other. Both Linacre and Wotton were serious students of Greek before they undertook medical studies, but once embarked upon medicine, both of them having studied at Padua, not only did they become especially conscious of the failings of medieval medicine in contrast to the classical, but the philosophical and literary aspects of Galen's writings must have caused them to retain a concern with Greek literature as a whole even though their primary consideration had come to be a single facet of the body of that literature. Furthermore, the scientific nature of their interest permitted no equivocation in their knowledge of the language. Translations of Galen or Hippocrates required an exactitude beyond that of purely literary treatises. But whatever the relevancy of such remarks, it is certainly of significance that among the first teachers of Greek in England were Linacre, Clement, and Wotton, all physicians, and for our present purpose as it relates to David Edwardes, it should be noted that two of these men, Clement and Wotton, were associated with Corpus Christi College. 8

In addition to the stress upon Greek studies which must inevitably have led Edwardes to the classical Greek writers upon medicine and conducted him along the pathway already marked out by Thomas Linacre, there were in the college certain possibly more direct influences towards an interest in medicine which have already been alluded to. In short, John Clement, the

early lecturer of Greek ^[26] was a physician and friend of Linacre as well as a fellow in the College of Physicians of London which Linacre had inaugurated in 1518, while still another student of medicine was Edward Wotton, Reader in Greek and later physician to Henry VIII, for whom Edwardes had briefly substituted.

Still another incentive toward medical study may have been a requirement in chapter 25 of the original statutes. In accordance with this all fellows of the college who held the degree of Master of Arts were required to assume holy orders, unless deputed to the study of medicine. It has been suggested that recipients of this exception were originally expected to attend to the medical needs of the other inmates of the college, ^[27] and it seems likely that Edwardes was one of these *medicinae deputati*.

Our next record indicates that he had removed to the University of Cambridge where in 1528-9, and upon payment of 3s. 4d. ^[28] and after lecturing publicly upon Galen's *De Differentiis Februm*, he was incepted in medicine with recognition of 'seven years study of medicine', presumably at Oxford. ^[29]

In his only known book, to be considered later, Edwardes informs us 9 that his first practice of medicine had been 'at Bristol, having left my ^[30] teachers only shortly before and begun to swim without any support', although it is not clear whether this represented a brief interlude between Oxford and Cambridge or after he had received his degree of Doctor of Medicine. Whatever the case may have been, the few remaining autobiographical references are to his practice in and around Cambridge. As a member of the Faculty of Medicine, it is possible that Edwardes was criticized for devoting an excessive amount of time to his private practice, since in 1530-1 permission was granted him to be excused from a statutory ^[31] requirement of attendance at 'all congregations, masses and exequies'. Nevertheless he participated in the examinations of at least two students, one in 1537-8 ^[32] and the other in 1540-1. ^[33]

Edwardes's little book, to which reference was made above, was published in London in 1532 [O.S.] by Robert Redman. It is composed of two treatises of which the first, entitled *On Symptoms and Prognostications* (*De Indiciis et Praecognitionibus*), deals with uroscopy and medical prognostication, and since it represents merely the continuation of a medieval tradition it is of little importance except, as has been said, for its few autobiographical details. In his practice of medicine Edwardes appears to have represented, as we might expect, a combination of the old and the new. While giving support to uroscopy and displaying some sympathy toward folk medicine, he also gave allegiance to Hippocrates and Galen, and like his continental colleagues of this period he was not averse to the introduction of a word or even several lines of Greek into his text, so indicating his enthusiasm for and his ties with the classical revival. Furthermore, he was certainly one of the first English physicians to appreciate Linacre and terms him 'the most learned physician of his age'.^[34]

10

The second treatise, *A Brief but Excellent Introduction to Anatomy* (*In Anatomicen Introductio Luculenta et Brevis*), is, as has been mentioned, the first work published in England which was devoted solely to anatomy, and therefore despite its brevity it deserves some consideration in the general history of medicine and even greater consideration in that of English medicine. Turning our attention now to this treatise on anatomy it should be first noted that although printed in the same volume with the work on medical symptoms and sharing a common title-page with that work, the treatise on anatomy has a separate dedication to Henry Howard (1517?-1547), Earl of Surrey. It had been at the request of Henry VIII that this young nobleman took up residence at Windsor and lived there from 1530 to 1532 as the companion of Henry's son, the Duke of Richmond. Since Edwardes had dedicated the first treatise to the Duke of Richmond on 21 December 1532, it is not difficult to comprehend his choice of the duke's companion for the second dedication which bears the date 1 January 1532, or, according to the Gregorian calendar, 1533. There is nothing remarkable about this latter dedication, which contains the usual flattery, except for the final passage. There the author remarked upon the ignorance of anatomy among physicians, sometimes with lethal results. He recognized that the subject of anatomy was a difficult one, hence his treatise has been written

with brevity and clarity. Later, as he promised, if opportunity were to be granted to him he would write a more elaborate work.

Hereafter, if God permit, I shall compose a complete book of anatomy in which I shall further the opinions of all the learned, to which my own opinion will be added. I could have done this at present but not, 11 however, with the same effort or with the form of an introduction preserved. It remains that this little book, which we have enlisted in the service of the commonwealth, may be pleasing to you, for it recognizes the existence of those very few unlearned physicians by whose mistakes many perish, from which this fact will be gathered, that no parts of the body should be unknown to physicians.

This promise of a more extensive work in which the author was to include his independent anatomical observations, presumably based on further human dissection, appears not to have been fulfilled or, at any rate, there is no record of any such later and more extended anatomical treatise by Edwardes.

The text of this *Introduction to Anatomy* fills no more than fifteen small pages, and its very brevity must have made it virtually useless; even the author says that it 'is indeed a slight work'. The plan of presentation is that which had been popularized by Mundinus and was required by the relative speeds with which the different parts of the body succumbed to putrefaction during the course of dissection. Thus Edwardes first describes the lower venter, that is, the abdomen, abdominal cavity, and pelvis, next the thorax, and finally the brain and nervous system. Within his very brief presentation no mention is made of the extremities while, relative to the limits of the discussion, a preponderance of attention has been devoted to what were considered the organs of nutrition and blood manufacture.

The anatomical nomenclature is mildly astonishing, especially when one considers the time and place of composition. But if one considers that Edwardes was sufficiently learned in Greek to act as Reader in Greek at his college for a short period, it will not be too amazing to find him somewhat scornful of the terms employed by those he calls 'Barbarians', that is, 12 the European school influenced by Moslem medical writers, chiefly through the *Canon* of Avicenna, which employed an anatomical

terminology drawn from Latin and from curious hybrid forms partly Latin, partly Greek, partly Arabic and in some parts from Hebrew. Edwardes, on the contrary, employs classical Greek terminology as, for example, omentum rather than the medieval *zirbus* and mesenteric in preference to *meseraic*. In so far as his description extends, his nomenclature is as 'modern', if not more so, than that of some of the more learned anatomists on the Continent. Yet, while his vocabulary may be more modern his anatomy is not. Indeed, in the introduction he remarked, as has been mentioned, that in the future he hoped to write a more extensive work 'to which my own opinion will be added'. By implication, then, in this first brief treatise he had drawn upon earlier authorities, and while we might expect that this student of Greek would turn to Galen and Hippocrates this is true only in part. The liver as he describes it is medieval, the three-chambered heart is Aristotelian, derived from those 'Barbarians' he scorned.

While the treatise is noteworthy as the first work written in England solely devoted to anatomy, the text intrinsically is of little further value except for one statement referring to the emulgent, or renal veins. 'In the body of that one whom we dissected very recently the left branch had a higher place of origin. Very often, however, the opposite occurs, so that the right emulgent vein is carried higher in the body.' Here we have the first reference to human dissection in England, in which, moreover, the anatomist observing through his own eyes rather than those of past authorities, noted a variation from the commonly given description of the emulgent veins, a description derived from Galen's anatomical studies on animals.

Little more can be said about Edwardes. He seems to have died about 13
[35] 1542, and perhaps this explains why the larger work was never to be published. Perhaps, had he remained at Oxford, he might have established an anatomical tradition, and so provided the influence which his book was not to have. Today only one copy of this little treatise is known, that in the library of the British Museum, and no consideration appears to have been paid to it from Edwardes's day to the present. However, its virtual extinction was not the result of hard usage by students such as that which determined the almost complete annihilation of Vesalius' *Tabulae Anatomicae*. As has been said, no contemporary mentioned Edwardes, despite the fact that his book was published in London. The edition must

have been a small one, and copies were not likely to have been preserved as other and better works on anatomy began to be imported from the Continent.

Meanwhile the universities continued their drowsy course so far unaffected in any way by the efforts of an alumnus of one of them. The barber-surgeons and surgeons appear to have been equally unproductive of anything new, still leaning upon earlier continental writers. Yet a few individuals recognized the need for improvement. Well before the surgeons of England received official encouragement for anatomical study the surgeons of Edinburgh had asked for and obtained bodies for dissection. On 1 July 1505 the magistrates of Edinburgh granted a Seal of Cause to the Guild of Surgeons and Barbers, and this was confirmed by James IV on 13 October 1506. Among the clauses regulating the practice of the barbers and the surgeons is one giving them the body of one felon each year for an anatomy:

... and that we may have anis [once] in the yeir ane condampnit man efter he be deid to mak antomell of, quhairthraw we may haif 14 experience, ilk ane to instruct vtheris ... and that na barbour, maister nor seruand, within this burgh hantt [practise] vse nor exerce the craft of Surregenrie without he be expert and knaw perfytelie the thingis [36] abouewritten.

Edinburgh, therefore, was the cradle of anatomical study in the British Isles. In England Thomas Linacre had founded the College of Physicians of London in 1518 with the idea of its being a select body of physicians to raise medical standards and maintain them through its power of licensing to practice. The need of more modern surgical texts was indicated by the publication in 1525 of a translation of the work of the late fifteenth-century German surgeon, Hieronymus Brunschwig, which contained a brief section on anatomy, but there appears to have been no attempt to produce a new and up-to-date surgery in England. The fact was that the more advanced books from continental Europe proceeded to smother any continuance of independent native efforts, and in the field of anatomy this makes the early appearance of David Edwardes's little treatise an astonishing chronological anomaly in the history of English anatomical writing. The importance of

anatomy was now to be recognized, but it would be a long time before another native English treatise on the subject was published.

The introduction of the officially recognized, and even encouraged, study of human anatomy into England was the result of influences brought to bear from several sources: the desire of King Henry VIII to improve the practice of medicine and surgery in England and possibly, too, with thoughts for a higher quality of military surgery; and the desire, as well, of some of the more thoughtful surgeons, of whom Thomas Vicary was probably one. So it was that in 1540 the Company of Barbers was united with the Fraternity of Surgeons to form what was called the United Company of Barber-Surgeons of which Thomas Vicary was named Master in 1541, an event handsomely commemorated in a painting commissioned from Hans Holbein the younger. 15

In the Charter by which the union was officially sanctioned, a statement is to be found which was to be of particular importance to the advancement of anatomical knowledge:

the sayd maysters or governours of the mistery and comminaltie of barbours and surgeons of London, and their successours yerely for ever after their sad discrecions at their free liberte and pleasure shal and maie have and take without contradiction foure persons condemned adiudged and put to deathe for feloni by the due order of the kynges lawe of thys realme for anatomies without any further sute or labour to be made to the kynges highnes his heyres or successours for the same. And to make incision of the same deade bodies or otherwyse to order the same after their said discrecions at their pleasures for their further and better knowlage instruction insight learnyng and experience in the sayd scyence or facultie of surgery. [37]

It is of interest to note that very soon after the Charter had been granted, Thomas Vicary approached the Lord Mayor and Aldermen of London to make sure that the Barber-Surgeons should receive the bodies of the felons for anatomical study. It would seem that the Court of Aldermen were not

sure how they should direct their Sheriffs, for the Minutes of the Court for 14 December 1540 state:

... Item, Master Laxton & Master Bowes, Shreves of this Citye, prayed the Advyse of this howse for & concernyng the Delyuerye ouer of one of the dedde bodyes of the Felons of late condempned to dethe within this Citye, And requyred of the seyde Master Shreves by Master Vycary & other the surgeons of this Citye for Annotamye, Accordyng to the fourme of an Acte of parlyament thereof lately made. And Agreyd that the same Acte be first seen & then Master Shreves to work ther after. 16

[\[39\]](#)

With human dissection material assured, the United Company proceeded to appoint a Reader of Anatomy, the first perhaps being Thomas Vicary, and although the intervening records of the company are not complete, it is known that in 1546 Dr. John Caius, lately returned from Padua, where he had been acquainted with and even lived for a time with the celebrated anatomist Andreas Vesalius, was appointed and held the position of Reader of Anatomy for the next seventeen years. In his brief autobiography Caius refers to these dissections which he performed ‘for almost twenty years’, and adds, ‘By the wish of the most illustrious prince Henry VIII, King of England, I performed them in London before the surgeons; among the

physicians at that time there was no dissection.’ [\[40\]](#) It may be assumed, however, that by ‘physicians’ Caius was referring to those of London rather than to those of the universities. Nevertheless, his remark helps to explain the lack of anatomical works which might have competed with those of the Continent. The physicians, although better trained in languages than the surgeons and, we may assume, literary exposition, were as yet not interested in the subject of anatomy.

Nevertheless it does seem somewhat incredible that the physicians were so late in taking up the practice of human dissection. While it is always dangerous to exceed the limits of evidence, this peculiar situation in regard to the College of Physicians of London requires that attention be called to a statute of the college reproduced by Munk who gives it the date 1569-70.

[\[41\]](#)

According to this authority, the terms employed in the statute,

reproduced below in translation, suggest that human anatomical dissection was already being employed by the physicians of the college at the time, although it seems impossible to determine whether or not the reference is to a period earlier than 1565 when Elizabeth granted them four bodies annually for anatomical purposes. However, it seems unlikely that the college, which was so concerned with the enforcement of laws concerning medicine would itself perform an illegal action and therefore that Elizabeth's grant to the college most likely introduced it to human dissection. Furthermore, one wonders just how frequently the college employed its new right, and in this respect it is interesting to note that there is no reference either to Elizabeth's grant or to any dissection at all in the *Annals* of the college as written by John Caius.

Although the study of human anatomy was now officially recognized and regularly pursued, at least in London, it would be incorrect to believe that native English anatomical writings would be forthcoming to continue the course modestly established by David Edwardes. The apathy or even hostility of physicians toward anatomical studies was an obstacle experienced earlier on the Continent and referred to by Vesalius who contributed no small share to the growth of anatomy's respectability in the eyes of physicians. However, the time lag between the Continent and England had resulted in a disregard of anatomical studies by English physicians at the very times when continental physicians had begun to interest themselves in the subject and publish anatomical studies. As a result it was inevitable that for such Englishmen as were interested in anatomy it was easier to import the more advanced and elaborate continental texts, and dependence on such alien works was for long to be the regular pattern. But even with these advanced, contemporary works available, the practice continued among the surgeons of republishing old and obsolete anatomico-surgical treatises of late medieval times. If such a practice was dictated by an elementary knowledge, certainly the continuance of it would not lead to any development.

In 1544 a Flemish engraver named Thomas Lambrit, better known under his pseudonym of Geminus, engraved on copper a series of anatomical figures plagiarized from the *Fabrica* and *Epitome* of Vesalius. Geminus displayed

the plates, which are of considerable artistic merit, indeed, the first of high quality to be produced in England, to King Henry VIII. That monarch, aware of the need of anatomical books to bolster the anatomical teaching now in progress, urged Geminus to publish his engravings. Never one to scorn the chance of gain, Geminus proceeded to follow this royal advice in the succeeding year (1545) and added to his plates a dedication to the king

and the text of Vesalius' *Epitome*.^[44] For some peculiar reason the completely innocent John Caius has occasionally been blamed as the impetus to this plagiarized publication despite the fact that Geminus states plainly in his preface that Henry VIII was responsible for his decision to publish.

While the illustrations plagiarized from Vesalius may have been of some pedagogical value, the text of the *Epitome* certainly was no anatomical manual, and the fact that it was in Latin, which many if not most of the surgeons could not read, gave it even less value.

It was perhaps at least partly for these reasons that Thomas Vicary appears to have issued in 1548 an anatomical text in English entitled *A Profitable Treatise of the Anatomie of Mans Body*. No copy of it is known to exist today, and its existence is realized only through mention of it on the title-page of an edition published in 1577 by the surgeons of St. Bartholomew's Hospital and a reference to it in 1565 by another surgeon, John Halle, 19 who refers to Vicary as 'the firste that euer wrote a treatyse of Anatomye in English (to the profite of his brethren chirurgiens and the

^[45] helpe of younge studientes) as farre as I can learne'. However, to refer to the 'profite' and 'helpe' to be obtained from Vicary's treatise is to reveal the deplorable state of anatomical studies in England at the time and to cause one to wonder if Halle had read by way of contrast the continental writings of that period. It seems very likely that what has been termed Vicary's anatomy was nothing other than a copy of a manuscript, presently in the Wellcome Historical Medical Library in London, dated 1392 and merely a compilation of Lanfranc, Henri de Mondeville, and Gui de Chauliac, the most recent of them dead in 1367. Thus not only was Vicary's work not based upon dissection, except for a secondhand account of crude

fourteenth-century autopsy, but it represented a definite case of retrogression.

The next anatomical publication in England was a new edition in 1553 of Geminus's plagiarized anatomical plates, but this time with an English text by Nicholas Udall, best known as the author of the first important English comedy, *Ralph Roister Doister*, and utterly lacking in knowledge of anatomy. In consequence one may correctly hazard that this work, published with commercial rather than pedagogical motives, would not contribute much to knowledge of anatomy in England, even though the text was now in English. It is true that Vesalius' descriptions of his illustrations were put into English, the first translation into English of any portion of the *Fabrica*, but the text which now replaced the *Epitome* of the earlier edition of 1545, like Vicary's work, is predominantly indebted to that same fourteenth-century manuscript compiled from the writings of late 20 medieval surgeons. Finally, the sheets of this work were reissued in 1559 with a new preface written by Richard Eden which aimed to delude the public into the belief that the publication had been revised.

About this time, too, a small series of anatomical fugitive sheets with superimposed flaps made their appearance in England. One, at least, had two leaves of English text to explain the woodcut and is nearly always discovered bound into the 1559 reissue of Geminus's book. The fugitive sheets, like their continental predecessors and followers, added very little to anatomical knowledge and must have been for popular consumption.

If we turn now for a moment to give consideration to continental activity during the same period, there is no difficulty in observing the superiority of publications abroad. In 1543 the *Fabrica* of Vesalius was published, in 1545 the *De Dissectione* of Rivière and Estienne, in 1555 the revised and much improved second edition of the *Fabrica*, in 1556 *Composicion del Cuerpo Humano* of Valverde, and in 1559 the *De Re Anatomica* of Colombo. It is little wonder that these foreign texts overwhelmed the English market and prevented any initiative which might have led to the publication of any but the most rudimentary manuals, presuming that there was in England anyone who had pursued the study of anatomy sufficiently to be in a position to compete with the continental authorities. On the other hand, the superiority

of the foreign publications owed part of that superiority to the fact that they were the work of much better educated physicians who had undertaken the study of anatomy, whereas in England the subject was yet very largely under the control of the less learned and less articulate surgeons who thought of anatomy more as a limited body of technical information required for surgery rather than a field of knowledge to be studied for 21 itself and capable of indefinite expansion. David Edwardes had sought to set medicine on the right course, but to no avail. While in time the Faculties of Medicine in the two universities would pay some lip-service to anatomy, yet some considerable time was to pass before they became genuinely interested in the subject.

In 1549 a royal examination of the Oxford statutes led to a declaration that they were ‘antiquated, semi-barbarous and obscure’, and new ones were substituted. In regard to medicine it was declared that before receiving the degree of Bachelor of Medicine the student must see two anatomical dissections, and himself perform two dissections before receiving his licence to practice. Before receiving the degree of Doctor of Medicine he

[46] was required to observe two or three more dissections. This, however, seems more likely to have been the ideal than the reality and echoes a similar but normally unfulfilled requirement in fifteenth-century Paris. It is more likely that the frequency with which anatomy was conducted at Oxford would have depended upon the particular interest of the Professor of Medicine, such as Walter Bayley (1529-93) who became Regius Professor of Medicine in 1561 and who at his death left his ‘skeleton of bones in

[47] Oxford’ to his successor in the chair. However, no Reader in Anatomy was appointed at Oxford until 1624. Indeed, the founder of the readership, Richard Tomlins, recognized the situation in his grant by noting that the study of anatomy was

more particularly necessary for the faculties and Artes of Phisicke and Chirurgery, the perfection whereof doth much avayle to the safety health and comfort of the whole Commonwealth in the conservation of their persons: And that there is as yet in neither of the Vniversities of this Kingdome (thoughe otherwise the most florissning of the whole 22

Christian world) any such Anatomy Lecture founded or established. ^[48]

If we may believe John Caius, writing after the middle of the century, the first early enthusiasm for Greek studies had worn off among physicians. Caius, himself a very competent Grecist, wrote in advocacy of the study of Greek medicine in the Greek language, that

as each is more capable in his own tongue so he is consistent and always remains himself which contributes much to clarity, since each tongue has its own idioms and inexpressible terms which when translated do not retain the same emphasis or a like grace. In short, translators some times do not understand certain things, elsewhere they fall asleep, do not retain exactness of diction, restrain freedom, and since we are all human and so desirous of variety, from time to time they slip so that not only may there be obscurity but even ambiguity.

Nevertheless, wrote Caius, in his day ‘everyone turns to the Latin editions and no one touches the Greek’. ^[49]

It is certainly true that after that first generation of men like Linacre, there was little interest in England in the original language of Galen and Hippocrates. The surgeons, certainly, knew no Greek, and the physicians were not interested in anatomy. There was to be little controversy, therefore, as to the meaning of any of Galen’s anatomical terms and less likelihood of investigating and disputing Galenic assertions. Acceptance without demur of the translation was a long step toward unquestioned acceptance of the content of the original. Hence it appears that by the middle of the sixteenth century the authority of Galen in Latin dress, or of his commentators, was not very likely to be opposed. On the Continent it had been instances of questions and opposition which had brought about anatomical advancement by resort to the only arbiter of doubts and questions, that is, the cadaver. 23

With conditions as they have been portrayed it is no wonder, therefore, that little initiative was displayed in England. The most popular of the foreign works in England, as on the Continent, appears to have been the *De Re Anatomica* of Colombo which held its position until well after the opening

of the seventeenth century. It was excellent for its time, not certainly the equal of the *Fabrica*, but on the other hand much cheaper to purchase, less bulky to hold, and not so detailed as to be confusing. It was probably this particular work in its several editions which more than any other prevented the appearance of a native English anatomical text.

In 1578 John Banister published a book entitled *The Historie of Man, sucked from the Sappe of the most approued Anathomistes*. The title indicates the character of the work, drawn from continental authorities, and especially from Colombo, despite the fact that Banister was Reader in Anatomy to the United Company and therefore in a position to undertake independent researches. Indeed, a contemporary painting shows Banister in his capacity as Reader standing beside an open copy of Colombo's *De Re Anatomica*.^[50]

It is clearly apparent that English anatomy in the Tudor period remained far behind that of the Continent, at least on the basis of such books as were published in England, and thereby renders that modest but early effort of David Edwardes all the more curious.

Edwardes, it must be recalled, had presented his brief treatise in the same form which was being employed on the Continent, and we may assume that it represented his method. What he did was to ignore medieval writers and return directly to Galen, the supreme authority of that age, the 'Prince of Physicians'. Coupled with this, he had begun to dissect, first, it may be assumed, for better comprehension of Galen but ultimately by Edwardes or his successors, discrepancies between the text of Galen and the observed anatomy would at once have indicated the classic error and the path to knowledge. Such was the course of continental development, but English anatomy of the period was faced by an insurmountable obstacle.

Whereas the medical faculties of continental universities came to accept anatomy, such was not to be the case with English medicine until well into the seventeenth century. As a result, anatomy was not an end in itself but rather a limited field of knowledge learned in so far as it might be usefully applied in surgery.

There were, of course, some Englishmen whose training and knowledge were superior to the quality demonstrated in English texts, men who had had Paduan training such as Caius and Harvey. But even Caius remained a Galenist when continental anatomy had become Vesalian, and Harvey, despite his thoroughly scientific attitude in respect to physiology, remained very conservative in his approach to purely anatomical problems, seeking authority not only in Galen but in the even more ancient Aristotle.

Under these conditions it seems remarkable that such great contributions were made to physiology in seventeenth-century England. The contributions of Harvey, Boyle, Hooke, and Lower form an amazing contrast to the static and even retrograde position of anatomy in the preceding century. In 1565 John Halle, a distinguished surgeon, published his *Anatomy or Dissection of the Body of Man* which was largely a translation of the surgery of Guido Lanfranc who died in 1315, yet fifty-one years later Harvey had arrived at the circulation of the blood. [\[51\]](#)

NOTES TO THE INTRODUCTION

- [1] Maurice Davidson, *Medicine in Oxford*, Oxford, 1953, pp. 15 ff.
- [2] H. D. Rolleston, *The Cambridge Medical School*, Cambridge, 1932, pp. 1 ff.
- [3] J. F. South, *Memorials of the Craft of Surgery in England*, ed. D'Arcy Power, London, 1886, pp. 14-15; Austin T. Young, *The Annals of the Barber-Surgeons of London*, London, 1890, p. 24.
- [4] South, op. cit., pp. 15-18.
- [5] Ibid., pp. 20 ff; Young, op. cit., pp. 40 ff.
- [6] South, op. cit., pp. 81 ff.
- [7] Ernest Wickersheimer, 'Les premières dissections à la Faculté de Médecine de Paris', *Bulletin de la Société de l'Histoire de Paris et de l'Ile-de-France*, 1910, xxxvii. 162-3.
- [8] *Commentaires de la Faculté de Médecine de l'Université de Paris (1395-1516)*, Paris, 1915, p. 286.

- [9] *Cronica Fratris Salimbene* (Monumenta Germaniae Historica: Scriptores), Hanover, 1905-13, p. 613.
- [10] Robert von Töply, in Puschmann, *Handbuch der Geschichte der Medizin*, Jena, 1903, ii. 199.
- [11] Ibid., p. 201.
- [12] Ibid., p. 209.
- [13] M. Roth, *Andreas Vesalius Bruxellensis*, Berlin, 1892, p. 13.
- [14] Töply, loc. cit., p. 212.
- [15] Ibid.
- [16] Montagu Burrow's, 'Memoir of William Grocyn', *Collectanea, Second Series* (Oxford Historical Society), Oxford, 1890, pp. 332 ff.
- [17] J. N. Johnson, *The Life of Thomas Linacre*, London, 1835, pp. 1-12.
- [18] G. B. Parks, *The English Traveller to Italy. The Middle Ages (to 1525)*, Stanford, Calif., 1955, pp. 457-60.
- [19] R. J. Mitchell, 'Thomas Linacre in Italy', *English Historical Review*, 1935, l. 696.

[20] This sequence was followed in Paris where in particular Guinther of Andernach and Jacobus Sylvius were proceeding from their study of Galen's medical writings to those of an anatomical nature.

[21] Thomas Fowler, *The History of Corpus Christi College*, Oxford, 1893, p. 381; *Register of the University of Oxford*, ed. Boase, Oxford, 1885, ii. 128, where he is mentioned as 'David Edwardys, disciple of the dyalectic art'.

[22] Ibid.

[23] Fowler, op. cit., pp. 58 and n., 85 n., 369 and n.

[24] *Register*, p. 128.

[25] Fowler, op. cit., p. 370 n.

[26] Ibid., p. 369.

[27] Ibid., p. 372.

[28] *Grace Book B*, ed. Mary Bateson, Cambridge, 1905, pt. ii, pp. 148, 150.

[29] *Grace Book Γ*, ed. William George Searle, Cambridge, 1908, p. 242.

[30] *De Indiciis et Praecognitionibus*, London, 1532, Ei^r.

[31] *Grace Book* Γ, p. 254.

[32] *Ibid.*, p. 326.

[33] *Ibid.*, p. 353.

[34] *De Indiciis et Praecognitionibus*, C₃^r.

[35] There is record of the probate of his will in that year in the Vice-Chancellor's Court in the University of Cambridge with mention of his wife Alice. The actual will, however, appears to be no longer in existence. Information kindly supplied by Miss H. E. Peek, Archivist of the University of Cambridge.

[36] J. D. Comrie, *History of Scottish Medicine*, London, 1932.

[37] *The Paintings of Hans Holbein*, ed. Ganz, London, 1956, nos. 218, 219.

[38] Young, *op. cit.*, p. 588.

[39] Guildhall, Repertory 10, f. 186, 14 Dec. 1540.

[40] *De Libris Propriis*, p. 90, in *The Works of John Caius*, M.D., ed. Venn, Cambridge, 1912.

27

[41] William Munk, *The Roll of the Royal College of Physicians of London*, London, 1878, iii. 351. The statute is cited from Goodall's

MS., *On College Affairs*, pp. 55-56: ‘Among our elders the Anatomical Lecture was considered of such importance that according to everyone’s recollection very few Fellows sought to be excused from that duty except for very grave reasons. However, lest it happen that frequent dispensations of that sort should become usual and customary and thence, so it was feared, lest such a useful institution should gradually perish, they decided to prevent it through the statutes, by slight penalties in the beginning and afterward increased and more severe according to the danger. We desiring to follow their prudent regulation, lest hereafter we admit Fellows into the Society influenced by a like hope of always declining this duty and not giving their attention seriously to that task: We establish and Order that for those refusing the duty of the ordinary anatomical lecture and wishing to be released wholly from that duty, the penalty of paying the College twenty pounds, unless because of very serious obstacles approved by the President and a majority of the Fellows in plenary session. In cases of lesser importance in which there is not sought a continuing exemption but a deferment from lecturing for a time, we leave to the judgment of the President how far this ought to be granted to the applicants; but the deferment granted may not exceed seven months. In which case also we wish that deferment from the first lecture may not be granted in favor of the succeeding lecturer, but that he be held to observe the time ordered for him by the President, as if there were no such deferment.’

[42]

Charles Goodall, *The Royal College of Physicians of London*, London, 1684, pp. 34-37: ‘Elizabeth by the grace of God, Queen of England, France & Ireland, defender of the faith &c. Greetings to all those reached by the present letter. Our father of noble memory Henry VIII, formerly King of England, among certain other decrees for the well-being and usefulness of his kingdom of England, especially watching over the health of his subjects, through his Letters Patent instituted in perpetuity a College of certain grave men of medicine who practised medicine publicly in his City of London and its suburbs within seven miles of that city. In the name of the President of the College and the Fellowship of the faculty of medicine of London, he incorporated them in the corporate and political body, and he granted to the same

President and College of Fellowship aforesaid and to its successors diverse liberties and privileges. Our same father not only confirmed those Letters Patent and all things contained in them through his *Senatus Consultum* or Parliament held in the fourteenth and fifteenth years of his reign, but also he increased and amplified the same statute in many ways. Since our said father granted this pious design for the well-being of the commonwealth, assuredly day by day there will be manifestly great advancement if to the aforesaid President, College or Fellowship and their successors forever we grant what is especially necessary for those professing medicine, certain human bodies annually for dissection. Know that we, not only deservedly renewing the famous institution of our said father, but also considering the responsibility of our royal office to provide as much as possible for the assured health and security of our subjects, of our special grace and from our certain knowledge and genuine affection for our people, we grant presently and for our heirs and successors to the aforesaid President of the College or Fellowship of the aforesaid faculty of medicine of London, and their successors or assigns, that they may have and receive annually and forever in future times, at one time or at different times of the year, at the discretion, desire and liberty of the aforesaid President during the time of his existence and of his successors, one, two, three or four human bodies for dissection and anatomization, which have been condemned and executed according to the common law of this kingdom for theft, homicide or whatever felony, or have been condemned and executed according to the common law of this kingdom for theft, homicide or whatever felony within the County of Middlesex or within the aforesaid City of London or elsewhere within sixteen miles of the aforesaid City in whatever County.... And that it be permitted to the same President of the College and aforesaid Fellowship and their successors and whatever others of their assigns, professors or experts, to dissect and to divide the same bodies or otherwise according to their will and judgment, with that reverence which ought to be granted to human flesh, for the increment of knowledge of medicine and experiment of the same, and for the health of our liegemen without the contradiction of anyone. And this without rendering or paying any one any sum of money or any sums of money for the same. Provided always that when from time to time an anatomy of this sort has been undertaken

and completed that the aforesaid bodies be given funeral and burial at the expense of the President and his successors....

Westminster, 24 February, in the seventh year of our reign'

[43]

Caius was not only a confirmed Galenist, but with the passing years ever a more conservative and literal Galenist, and his anatomical lectures to the surgeons were described by Bullein in his *Little Dialogue* of 1579 as 'reveiling ... the hidden jewels and precious treasures of Cl. Galenus'. It seems likely that, whatever anatomical lectures were given in the College of Physicians, they must, at least for a time, have been of like character. In the *Annals* of the college as written by Caius we find that as late as the year 1559 a certain Joannes Geynes was subject to disciplinary action because of his assertion that Galen had been guilty of error. He was required to state that 'I Joannes Geynes confess that Galen did not err in those things for which I criticized him', *Annales a Collegio Condito*, pp. 53-54, in *The Works of John Caius*, ed. Venn, Cambridge, 1912. Such conservatism carried over to the study of anatomy would certainly have been detrimental to any advancement of knowledge.

29

[44]

Compendiosa totius Anatomie delineatio, aere exarata per Thomam Geminum.

[45]

Selected Writings of Sir D'Arcy Power, Oxford, 1931, p. 115.

[46]

H. M. Sinclair and A. H. T. Robb-Smith, *A Short History of Anatomical Teaching in Oxford*, Oxford, 1950, p. 10.

[47]

Ibid., p. 11.

[48]

Ibid.

[49]

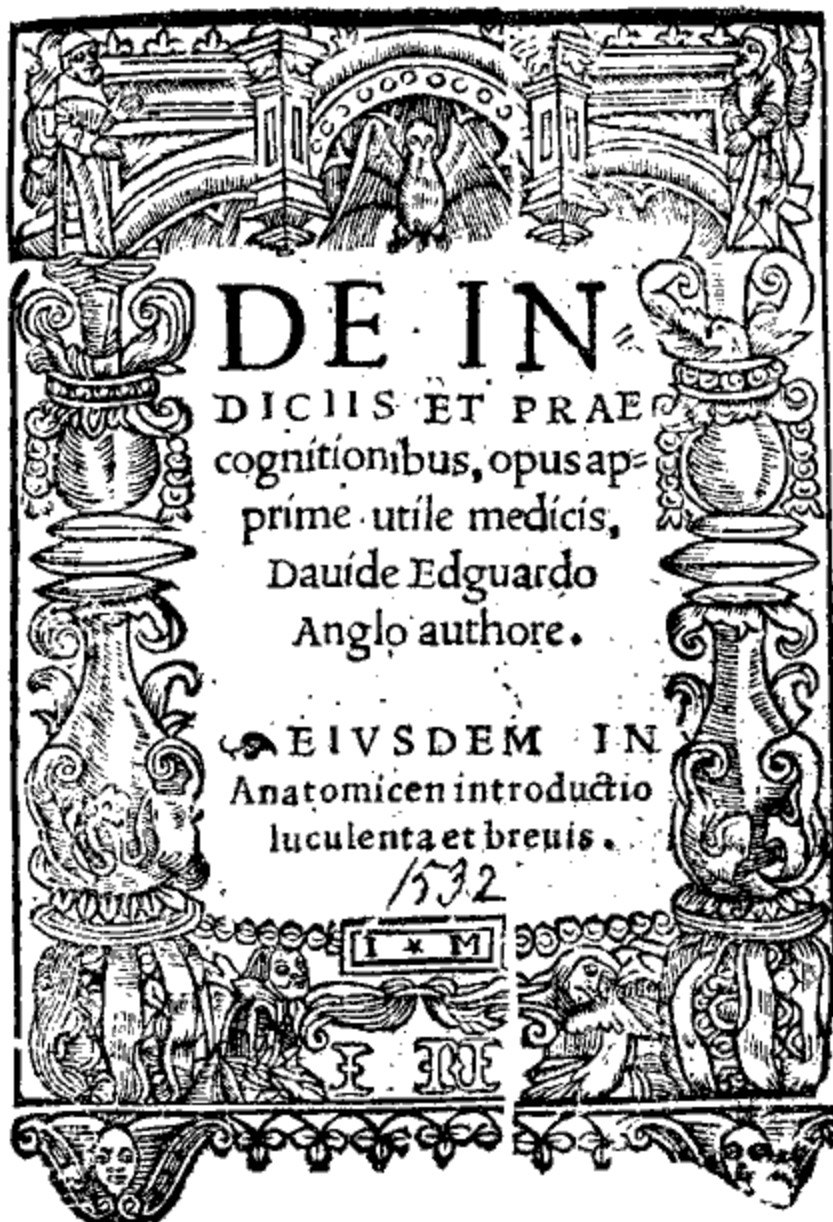
Caius, loc. cit., p. 104.

[50]

This portrait shows Banister giving the Visceral lecture at Barber-Surgeon's Hall in 1581; of small size and painted by an unknown artist on two pieces of paper joined down the middle, it is nevertheless sufficiently detailed for us to discover that Banister is using the octavo edition of Colombo's work printed in Paris in 1572. The portrait is now laid down in an album of anatomical drawings, also painted for Banister, which was formerly owned by William Hunter and is now preserved in the Hunterian Museum, Glasgow. The drawings consist of views of the skeleton, the superficial muscles, nerves and veins drawn in colour on a dark ground with some skill. Singer, in his *Evolution of Anatomy*, London, 1925, p. 174, suggests that the skeletal figures are probably the earliest prepared in England which were actually drawn from the bones. This could well be true, but Banister based his drawing of the nerves on a plate of Charles Estienne, 1545, and his figures of the superficial muscles and veins are possibly based on Valverde. Other relics of Banister can be seen at Cambridge. The University Library has a book-like casket containing a small ivory skeleton and the *écorché* figure of a man given to the library by Banister in 1591. King's College Library has a copy of *The Historie of Man* presented by the author in 1596.

[51]

Books printed on the Continent were freely available in England, and it could be argued that this was one reason why so few anatomical texts were published in the Tudor period. It has already been noted that Colombo's *De Re Anatomica* in the octavo edition of Paris, 1572, was used by Banister in his visceral lecture. This could well have been the text recommended to apprentices of Barber-Surgeon's Hall. Such imported books were, of course, published in Latin and were therefore suitable to the students of the College of Physicians and those of Oxford or of Cambridge. It seems likely that the students at Barber-Surgeons' Hall created a demand for more simple texts in the vernacular and this is surely the reason for the continued popularity of such books as Thomas Vicary's archaic text.



DE IN-
DICIIS ET PRAE
cognitionibus, opus ap-
prime utile medicis,
Dauidе Edgardo
Anglo authore.

EIVSDEM IN
Anatomicen introductio
luculenta et brevis.

1532

EXIMIO

AC ILLVSTRI D. HENRICO

Surrensi Comiti Da. Edguardus

medicus S. D.



VOTIES MIHI
in memoriā uenit Hen-
rice nobilium Comitum
decus, et quanto in ho-
nore fuerit tuus auus a-
pud Anglos omneis, cū ob insignem illā
uirtutē suam et fortunatos in rebus bel-
licis succēssus, tum ob prudentiam suam
minime uulgarem in administranda re
publica, dum uiueret: et quam dextere eti-
am his diebus quotidie gerantur res om-
nes tuo patri praeclarissimo, quaecunq; ad
nos Anglos pertinent: non possum satis
admirari genus istud tuum, non horum
adeo causa, ut quod et te in hac aetate cō-
stitutum, uideam, supra quam dici potest

EXIMIO

AC ILLVSTRI D. HENRICO
surrensi Comiti Da. Edguardus
medicus S. D.

QVOTIES MIHI

in memoriā uenit Hen-
rice nobilium Comitum
decus, et quanto in ho-
nore fuerit tuus auus a-
pud Anglos omneis, cū ob insignem illā
uirtutē suam et fortunatos in rebus bel-
licis succæssus, tum ob prudentiam suam
minime uulgarem in administranda re
publica, dum uiueret: et quam dextere eti-
am his diebus quotidie gerantur res om-
nes tuo patri præclarissimo, quæcunq ad
nos Anglos pertinent: non possum satis
admirari genus istud tuum, non horum
adeo caussa, ut quod et te in hac ætate cō-
stitutum, uideam, supra quam dici potest

in multis alijs adolescentibus, ad ea quæ te meliorem reddant tam serio animum appellere. Istud quidem ego haud scio, naturę ne illius beneficio ascribere debeat, è qua nobis editus es, an superis, qui et tu is bene uertunt, et Anglis nostris fauent. Vt cunq; est, reipublicę nostrę commodo fore speramus, quod factum est, atq; eo magis, quo tu diutius rebus bonis studueris. Ita namq; sequentem ætatem instructior adibis, et cõsuetudo interim bona tuum animum stabiliet, ne ad peiora in posterum facile decidas. Quanto uero magis et consilio ualebis, et prudentia, tanto meliorem sui gubernatorem habebit Norfolkia tuę stirpis credita, ubi patri succædes hæres prædiorum, tantoq; interea utilior Comes eris Surrensi populo tuo, ac tanto demum magis Anglis omnibus expetitus, ut reipublicę negotia suscipias, quæ omnia et honorum tibi incrementum futura sunt et tuorum omnium.

in multis alijs adolescentibus, ad ea quæ te meliorem reddant tam serio animum appellere. Istud quidem ego haud scio, naturæ ne illius beneficio ascribere debeā è qua nobis editus es, an superis, qui et tuis bene uertunt, et Anglis nostris fauent. Vt cumq̃ est, reipublicæ nostræ commodo fore speramus, quod factum est, atq̃ eo magis, quo tu diutius rebus bonis studueris. Ita namq̃ sequentem ætatem instructior adibis, et cōsuetudo interim bona tuum animum stabiliet, ne ad peiora in posterum facile decidas. Quanto uero magis et consilio ualebis, et prudentia, tanto meliorem sui gubernatorem habebit Norfolcia tuæ stirpi credita, ubi patri succædes hæres prædiorum, tantoq̃ interea utilior comes eris surrensi populo tuo, ac tanto demum magis Anglis omnibus expetitus, ut reipublicæ negotia suscipias, quæ omnia et honorum tibi incremento futura sunt et tuorum omnium

honestamento quæ omnia et certum est
consequi posse te, si ut coepisti iuuenileis
annos transiges. Vt magnam in te spem
reponimus, te et patris aui tui similem fu-
turum, quod et ingenium tuum et morum
grauitas talia nobis polliceatur. Ego tibi
et maximos successus precor et optima-
rum omnium rerum augmentum uber-
rimum. Atque ut hic annus totus ab initio
foelix tibi tuisque sit, iterum precor. Quo-
omine et hanc nostram in Anatomicen
introductionem tibi dedico. Vt enim
hæc artis medicæ pars omnibus comper-
ta non est, sic et quod difficillima nonnul-
la complectatur, facilem exigit instituti-
onem, qua lectores quasi manu ducantur
ad id, cui innituntur. Istud opus exiguum
quidem est, sed medicis et Chirurgis om-
nibus per quam utile, quod et plurima
paucis explicat. Nihilò obscuri, nihil af-
fectati continet, sed omnibus eorum in-
genijs expositissimum, qui nec tardi sunt,

honestamento quæ omnia et certum est consequi posse te, si ut cœpisti iuuenileis annos transiges. Vt magnam in te spem reponimus, te et patris aui tui similem futurum, quòd et ingeniū tuum et morum grauitas talia nobis polliceātur. Ego tibi et maximos succæssus precor et optimarum omnium rerum augmentum uberimum. Atq̃ ut hic annus totus ab inicio fœlix tibi tuisq̃ sit, iterum precor. Quo omine et hanc nostram in Anatomicen introductionem tibi dedico. Vt enim hæc artis medicæ pars omnibus comperita non est, sic et quod difficillima nonnulla complectatur, facilem exigit institutionem, qua lectores quasi manu ducantur ad id, cui innitūtur. Istud opus exiguum quidem est, sed medicis et chirurgis omnibus per quam utile, quod et plurima paucis explicat. Nihilo obscuri, nihil affectati continet, sed omnibus eorum ingenijs expositissimum, qui nec tardi sunt,

nec ad scientias inutiles . In quo , si quæ
discrepent à communibus medicorū sen-
tencijs, nemo miretur: quòd neq; doctis-
simi eadē his de rebus sentiant. Ego post
hac, si deus permittet, librum condam ab-
solutæ Anatomices, in quem doctissimo-
rum omnium opiniones colligam , quib;
et mea sententiā interponetur. Potui
et id iam facere , sed nec eodem tamen la-
bore, neq; seruato introductionis decoro.
Supereſt ut hic libellus tibi gratus ſis
quem in reipublicæ commodum cōscrip-
ſimus . Reddet enim pauciores indoctos
medicos, quorum uicio plurimi intereunt
à quo et hic fructus percipietur ,
ut nullæ corporis partes me-
dicis non ſint notiſ-
ſimæ. Vale. Can-
tabrigiæ, Cal.
Ianua-
rij.

nec ad scientias inutiles. In quo, si quæ
discrepent à communibus medicorū sen-
tencijs, nemo miretur: quòd neq̃ doctis-
simi eadē his de rebus sentiant. Ego post
hac, si deus permittet, librum condam ab-
solutæ Anatomices, in quem doctissimo-
rum omnium opiniones colligam, qui-
bus et mea sententia interponetur. Potui
et id iam facere, sed nec eodem tamen la-
bore, neq̃ seruato introductionis decoro.
Superest ut hic libellus tibi gratus sit
quem in reipublicæ commodum cōscrip-
simus. Reddet enim pauciores indoctos
medicos, quorum uicio plurimi intereunt
à quo et hic fructus percipietur,
ut nullæ corporis partes me-
dicis non sint notis-
sime. Vale. can-
tabrigiæ, cal.
ianua-
rij.

DAVIDIS

EDGVARDI ANGLI IN

Anatomicen introductio.



INFERIOR uenter totus (hinc enim humani corporis incipere dissectionem oportet, quod is locus ocysissime putrescat) à prima cute ad peritonæum Græcis ἐπιστάσιον, Barbaris Mirach appellatur cuius quidem hæ partes sunt.

CUTIS IN superficie quæ totum occupat corpus, sensus omnis expers. Cutis tenuissima superficiali cuti subiecta et subtenfa, sensibilis. Græci eam ὑπόδερμα dicunt. Pinguetudo quædam totum uentrem occupans, cuti sensili citra medium substrata.

NERVOSA et tenuis membrana

DAVIDIS

EDGVARDI ANGLI IN
Anatomicen introductio.

INFERIOR

uenter totus (hinc enim humani corporis incipere dissectionem oportet, quòd is locus ocyssime putrescat) à prima cute ad peritonæū Græcis ἐπιγάστριον, Barbaris Mirach appellatur cuius quidem hæ partes sunt.

CVTIS IN superficie quæ totum occupat corpus, sensus omnis expers. Cutis tenuissima superficiali cuti subiecta et subtensa, sensilis. Græci eam ὑποδερμα dicunt. Pinguetudo quædam totum uentrem occupans, cuti sensili citra medium substrata.

NERVOSA et tenuis membrana

INTRODUCTIO.

hæc continuo sequitur. Membrana rursus è musculis ortum habens huic statim subiungitur, ubi recta mox linea apparet in medio.

MUSCULI obliqui duo descendentes uersus inum uentrem his subiacent. Musculis obliquis ascendentibus sub his locus est. Recti duo musculi uendicant locum proximum. Atque infimi omnium sunt musculi transuersi. Octo igitur hi sunt quorum ferè singulis sunt suæ tunicae neruosaë quibus à se inuicem discriminantur.

SVBTENDITVR his aponeurosis siue potius membrana quædam spissa et tenax quam aliqui falso peritonæum appellant. Hactenus de Epigastrio et eius partibus.

CERTÈ peritonæum neruosa pars est, tactu mollis, tenacitatis mediocris, totum uentrem occupans, et aponeurosi si-

hæc continuo sequitur. Membrana rursus è musculis ortum habens huic statim subiungitur, ubi recta mox linea apparet in medio.

MVSCVLI obliqui duo descendentes uersus imum uentrem his subiacent. Musculis obliquis ascendentibus sub his locus est. Recti duo musculi uendicant locum proximum. Atq̃ infimi omnium sunt musculi transuersi. Octo igitur hi sunt quorum ferè singulis sunt suæ tunicæ neruosæ quibus à se inuicem discriminantur.

SVBTENDITVR his aponeurōsis siue potius membrana quædam spissa et tenax quam aliqui falsò peritonæum appellant. Hactenus de Epigastrio et eius partibus.

CERTÈ peritonæum neruosa pars est, tactu mollis, tenacitatis mediocris, totum uentrem occupans, et aponeurōsi si-

IN ANATOMICE N.

ue membranæ quam dixi subsidēs. Græci id illi nomen indiderunt. Barbari Siphac id uocant.

ZIRBUS siue omentum subperitonæo exporrigitur. Adipis quoddam genus Zirbus est, ex neruosis filis tenuibusq; neruorum substantia adiposa constans priore adipe minus crassū. Intestina plurima et imum uentriculum operit, et alimentī coctionem expedit.

INTESTINA à uentriculo exoriantur, è quibus quod rectum et longanum appellatur, omnium intestinorum infimum est, et siccum alui onus continet, et inter nates caput exerit, ut onus deponat. Colon illi continuatur, et in ascensu renem sinistram ambit, et ad uentriculi latera dextrorsum cædit. Quod Græci τυφλόν et μονόφθαλμον, Romani cæcum intestinum et unoculum uocant, colo accrescit, cuius unicus est meatus, al-

ue membranæ quam dixi subsidēs. Græci id illi nomen indiderunt. Barbari si-phacid uocant.

ZIRBVS siue omentum subperitonæo exporrigitur. Adipis quoddam genus zirbus est, ex neruosis filis tenuiq̃ neruorum substantia adiposa constās priore adipe minus crassū. Intestina plurima et imum uentriculum operit, et alimenti coctionem expedit.

INTESTINA à uentriculo exoruntur, è quibus quod rectum et longanum appellatur, omnium intestinorum infimum est, et siccum alui onus continet, et inter nates caput exerit, ut onus deponat. Colon illi continuatur, et in ascensu renem sinistrum ambit, et ad uentriculi latera dextrosum cædit. Quod Græci τυφλον et μονόφθαλμον, Romani cæcum intestinum et unoculum uocant, colo accrescit, cuius unicus est meatus, al-

IN ANATOMICE N

tera enim extremitate clauditur, ut coctio
ni subseruiat cōmodius, uentriculi cuius-
dam modo. Hinc igitur rei nomen. Atq;
intestina quidem crassiora tot sint.

I L E O N excipit cæcum, intestinum
in crebros intortum sinus a qua figura et
Græci nomen illi fecerunt $\pi α ρ α ῖ ῥ ἰ ε λ ι ε ς$
 $δ α$. i. ab inuoluendo, cuius morbus et
iliacus dicitur. Illi iejunum adheret. Hoc
ieiuni nomen corporum dissectores in-
testino dederunt ab euentu quod semper
inane reperitur, et nihil continere. Iecur
enim auulsit prius quicquid haberet in se
ieiunum. Assurgit supra hæc intestina
omnia, duodenum quod iejuno inferne,
superne Pyloro affigitur. Græcis $δ ω δ ε κ α$
 $δ ᾱ κ τ ῦ λ ο ρ$ uocatur a duodecim digitorū
mensura. Triā hæc substantiæ suæ ratios
ne appellantur gracilia intestina.

V E N T R I C V L V S sub transuerso
septo locatus est, cuius os superius in
quod

tera enim extremitate clauditur, ut coctioni subseruiat cōmodius, uentriculi cuiusdam modo. Hinc igitur rei nomen. Atq intestina quidem crassiora tot sint.

ILEON excipit cæcum, intestinum in crebros intortum sinus a qua figura et Græci nomen illi fecerunt παρα του ἐιλεῖσθαι .i. ab inuoluendo, cuius morbus et iliacus dicitur. Illi ieiumum adheret. Hoc ieiumi nomen corporum dissectores intestino dederunt ab euentu quòd semper inane reperiatur, et nihil continere. Iecur enim auulsit prius quicquid haberet in se ieiumum. Assurgit supra hæc intestina omnia, duodenum quod ieiuo inferne, superne Pyloro affigitur. Græcis δωδεκα δάκτυλον uocatur à duodecim digitorū mensura. Tria hæc substantiæ suæ ratione appellantur gracilia intestina.

VENTRICVLVS sub transuerso septo locatus est, cuius os superius in

INTRODVCTIO.

quod œsophagus terminatur, stomachus propriè appellatur, inferius per
 q̃ intestina alimentũ deriuatur πύλωρος
 dicitur.

LIEN RARAE substantiæ uiscus,
 uentriculo adiacet ad sinistrum latus
 et iecur ad dexterum Hypochondrium,
 hoc rotundum, ac quadam tenus lunare,
 illud oblongum, ac ueluti quadratum,
 Vtriusq; horum gibbosa pars ad inferiores
 costas pertinet. Quod in alterutro cõ-
 cauum est, id et uentriculo est proximum.
 Iecur sanguinẽ gignit. Lien eundẽ repur-
 gat ab atrabile. Inuauescit Lien cũ reli-
 qui corporis dispendio. Iecoris magnitu-
 do totius corporis compagi utilis est,
 quòd sanguinẽ et naturalẽ spiritũ summi-
 nistret ubertim. Iecur habet suas penus-
 las quos Græci λοβοίς nominant, inter-
 dum treis, interdum plureis, in cuius ca-
 uo et uessicula fellis prominet, qua san-

quod œsophagus terminatur, stomachus proprie appellatur, inferius per *quod* intestina alimentū deriuatur πυλωρος dicitur.

LIEN RARAE substantiæ uiscus,
uentriculo adiacet ad sinistrum latus
et iecur ad dexterum Hypochondrium,
hoc rotundum, ac quadam tenus lunare,
illud oblongum, ac ueluti quadratum.
Vtriusq̃ horum gibbosa pars ad inferiores costas pertinet. Quod in alterutro cōcauum est, id et uentriculo est proximum. Iecur sanguinē gignit. Lien eundē repurgat ab atrabile. Inuaugescit Lien cū reliqui corporis dispendio. Iecoris magnitudo totius corporis compagi utilis est, quòd sanguinē et naturalē spiritū sumministret ubertim. Iecur habet suas penulas quos Græci λοβούς nominant, interdum treis, interdum plureis, in cuius cauo et uessicula fellis prominet, qua san-

IN ANATOMICEN

guis à bile defecatus et purus euadit.
Cuius utiq; uesiculæ exhalatione et transpiratu inficiuntur nonnunquam duodenum et ieiunum, nonnunquam et pungi se sentiunt, si transpiratus maior sit et bilis mordacior.

À IECORIS concavo uena portæ oritur: multis exilibus iecoris uenis concurrentibus ex quibus ea una constat. E diuerso rursus in innumeras eadem spargitur partibus, uenarumq; immensam reddit multitudinem, quæ postea passim intestinis propè omnibus inseruntur, admixtis unâ membranulis adiposis, ut nutrimentalem substantiam iecori suppeditent in sanguinis generationem. Chylus namq; cibusq; à uentriculo statim ad intestina demittitur concedente exitu pyloro, ubi primum accæperit uenter quantum usibus suis sufficiat, et coctionem suam peragerit qui nisi et in sanguis

guis à bile defecatus et purus euadit.
Cuius utiq̃ uessiculæ exhalatione et transpiratu inficiuntur nonnunquam duodenum et ieiunum, nonnunquam et pungi se sentiunt, si transpiratus maior sit et bilis mordacior.

À IECORIS concauo uena portæ oritur: multis exilibus iecoris uenis concurrentibus ex quibus ea una constat. E diuerso rursus in innumeras eadem spargitur parteis, uenarumq̃ immensam reddit multitudinem, quæ postea passim intestinis propè omnibus inseruntur, admistis unâ membranulis adiposis, ut nutritiua substantiam iecori suppeditent in sanguinis generationem. Chilus namq̃ cibusq̃ à uentriculo statim ad intestina demittitur concædente exitū pyloro, ubi primum accæperit uenterculus quantum usibus suis sufficiat, et coctionem suam perægerit qui nisi et in sangui-

INTRODUCTIO.

nis naturam transmutandus sit, parum admodum in reliqui corporis nutritio- nem contulerit. Hunc ergo usum præ- stant numerosæ hæ uenulæ, ut optimum nutrimenti succū haud satis adhuc coctū interaneis exugant, et iecoris cauo man- dent, quo illic sanguis fiat. Quas nimis- rum uenulas et Meseraicas, et Mesenteria- cas Græco uocabulo nominare licebit. Latini eas lacteis uocant. Ad harum mu- nimen ne per ramificationis frequentiam ualentiore corporis motu earum quæuis distrahantur dilanientur ue, quo firmitus consent singulæ sibi uenulæ duodeno πᾶσκρασ adhæret, glandulosa scilicet ca- ro, quæ et καλλικρασ Græcis uocatur in- terdum.

SANGVIS meat à iecoris cōcauo, in quo paulo ante formatus est, ad gibbū iecoris, non qualis tamen omnino factus fuerit in cauo, sed syncerior et simplicior,

nis naturam transmutandus sit, parum admodum in reliqui corporis nutricionem contulerit. Hunc ergo usum præstant numerosæ hæ uenulæ, ut optimum nutrimenti succū haud satis adhuc coctū interaneis exugant, et iecoris cauo mandent, quo illic sanguis fiat. Quas nimirum uenulas et meseraicas, et mesentericas Græco uocabulo nominare licebit. Latini eas lacteis uocant. Ad harum munimen ne per ramificationis frequentiam ualentiore corporis motu earum quæuis distrahantur dilanientur ue, quo firmitus constant singulæ sibi uenulæ duodeno πανκρεας adhæret, glandulosa scilicet caro, quæ et καλλίκρεας Græcis uocatur interdum.

SANGVIS meat à iecoris cōcauo, in quo paulo ante formatus est, ad gibbū iecoris, non qualis tamen omnino factus fuerit in cauo, sed syncerior et simplicior,

IN ANATOMICEN

utrâq; bile ab eo secreta, et ad conceptacula sua transmissa, ut corpori salubriter alendo et gignēdis spiritibus inculpatior sit. A gibbo uero et in totum undiq; corpus porrigitur sanguis, per uenam cauā (Græcis κοιλὴ dicitur) et multiplices eius uenæ ramos. Hæc profecto uena reliquas omneis corporis uenas inagnitudine superat, et à iecoris oritur gibbo. A qua per mediam spinam descendere unus utrinq; ramos renes petit, alterutro ramo in palmi longitudinem protenso.

HI CONCAVAE uenæ ramī uenæ sunt emulgentes. Quem nouissime secūimus, illi leuus ramus in corpore alciozem exortus sui locum habebat. Sepissime tamen contra fit, ut emulgens dextera uena sublimius in corpus efferatur. His emulgentibus uenis natura utitur ad deferendam sanguinis aquositate et bilem à iecore ad renes. Totidem et ar-

utrâq bile ab eo secreta, et ad concæptacula sua transmissa, ut corpori salubriter alendo et gignēdis spiritibus inculpatior sit. À gibbo uero et in totum undiq corpus porrigitur sanguis, per uenam cauā (Græcis κοιλη dicitur) et multiplices eius uenæ ramos. Hæc profecto uena reliquas omneis corporis uenas inagnitudine superat, et à iecoris oritur gibbo. À qua per mediam spinam descendēte unus utrinq ramos renes petit, alterutro ramo in palmi longitudinem protenso.

HI CONCAVAE uenæ rami uenæ sunt emulgentes. Quem nouissime secuimus, illi leuus ramus in corpore alciolem exortus sui locum habebat. Sequissime tamen contra fit, ut emulgens dextera uena sublimius in corpus effertur. His emulgentibus uenis natura utitur ad deferendam sanguinis aquositatē et bilem à iecore ad renes. Totidem et ar-

INTRODVCTIO.

teriarum ramulí, eodem situ, et á magna Aorta arteria cauam uenam subeunte, in æquam longitudínem procurrunt in renes, sub emulgentibus uenís, bile et sanguine aquoso cor exonerantes, quibus et arteriarum emulgentium nomen est.

DESCENDVNT et á sinistris emulgentibus uena et arteria in sinistræ partis testem. Seminales ex sunt meatus sanguine et spiritu turgentes, foeminas in his contenta seminis materia procreat, quód humor sit aquosus et coctionem desyderet. Meatus seminales itidem arteria et uena á dexteris demittuntur in dexterum testem, uerum á uenæ cauæ et Aortæ arteriæ truncis excrecentes, ac proinde succus in eis minus aquosus, ac probe cōcoctus, maribus generandis aptior est. In his meatibus sanguis percoquitur, qui póst ad glandulosam testium carnē translatus, seminis formam acquirit.

teriarum ramuli, eodem situ, et á magna Aorta arteria cauam uenam subeunte, in æquam longitudinem procurrunt in renes, sub emulgentibus uenis, bile et sanguine aquoso cor exonerantes, quibus et arteriarum emulgentium nomen est.

DESCENDVNT et à sinistris emulgentibus uena et arteria in sinistræ partis testem. Seminales eæ sunt meatus sanguine et spiritu turgentes, fœminas in his contenta seminis materia procreat, quód humor sit aquosus et coctionem desyderet. Meatus seminales itidem arteria et uena à dexteris demittuntur in dexterum testem, uerum à uenæ cauæ et Aortæ arteriæ truncis excrescentes, ac proinde succus in eis minus aquosus, ac probe cōcoctus, maribus generandis aptior est. In his meatibus sanguis percoquitur, qui póst ad glandulosam testium carnē translatus, seminis formam acquirit.

IN ANATOMICEN

RENES solida et dura uiscera sunt, non sentientia, uis attractrix in eis pollet plurimum. Sanguinem ab aquositate ac bile purgant. Sed sanguinem retinent, ut quo alantur reliquum humorem exprimunt. Eis enim *δυστηριάζει* adnectuntur. i. urinarij meatus, candidi fistulosi, ac tensiles, quales nimirum ad uesicam pertinere dixeris et eius substantiæ confineis esse.

SEPTVM transuersum est membranosa quædam substantia, uitalia et naturalia membra intercurlans. Græcis *Διαφραγμα* dicitur. Interraneis uim expultricem firmat, spiritui destinatis membris inscribitur, fumidosq; uapores coercet ne cordis, aut cerebri, uiuidos spiritus offuscent. Cui superne affigitur nervosa tunica quæ Thoracem intrinsecus uestit, et pectoris costas statis intersticijs deligat, quam tunicam Græci *πλευρα* bona ex parte nominant, aliquando uero *ἰνέζον* et eisdem

RENES solida et dura uiscera sunt,
non sentientia, uis attractrix in eis pollet
plurimum. Sanguinem ab aquositate ac
bile purgant. Sed sanguinem retinent, ut
quo alantur reliquum humorem expri-
munt. Eis enim οὐρητῆρες adnectūtur .i.
urinarij meatus, candidi fistulosi, ac ten-
siles, quales nimirum ad uesticam pertine-
re dixeris et eius substantiæ confineis esse.

SEPTVM transuersū est mēbranosa
quædam substantia, uitalia et natura-
lia membra intercursans. Græcis διάφραγ-
μα dicitur. Interraneis uim expultricem
firmat, spiritui destinatis membris inscri-
bitur, fumidosq̃ uapores coërcet ne cor-
dis, aut cerebri, uiuidos spiritus offuscent.
Cui superne affigitur neruosa tunica quæ
Thoracem intrinsecus uestit, et pectoris
costas statis intersticijs deligat, quam tu-
nicam Græci πλεῦραν bona ex parte no-
mināt, aliquando uero ὑποζωμα ijsdem

INTRODVCTIO,

uocatur. Huius inflammatione fit Pleuritis, morbi nomine à tunica ducto.

APLEVR A iuxta spinam nascitur et membrana pulmones et intimū Thoracem æquis portionibus per media distinguens. Mediastinum uulgo appellatur, pulmonibus tanto commodo insitum ut alterius pulmonis uicium alteri facile ex eo non communicetur. Certè pulmones in medio pectoris palacio habitant, cordis et cerebri spiritus recreant, calorem attemperant, et præfocationis periculum auertunt, quibus et suæ sunt penulle perinde atq; iocinori. Habent et cor perpetuo in quibusdam ueluti amplexibus blandissimarum nutricum more, et qualitatū quendam concentum acciunt, quo singulas corporis particulas demulceant, et uegetas faciant. E mediastini parte illa quæ medios habet pulmones, profert se membrana egregie spissa,

uocatur. Huius inflammatione fit pleu-
relis, morbi nomine à tunica ducto.

A PLEVRA iuxta spinam nascitur
et membrana pulmones et intimū
Thoracem æquis portionibus per media
distinguens. Mediastinum uulgo appel-
latur, pulmonibus tanto commodo infi-
tum ut alterius pulmonis uicium alteri fa-
cile ex eo non communicetur. Certè pul-
mones in medio pectoris palacio habi-
tant, cordis et cerebri spiritus recreant, ca-
lorem attemperant, et præfocationis peri-
culum auertunt, quibus et suæ sunt pe-
nulle perinde atq̃ iocinori. Habent et cor
perpetuo in quibusdam ueluti amplexi-
bus blandissimarum nutricum more, et
qualitatum quendam concentum acci-
nunt, quo singulas corporis particulas de-
mulceant, et uegetas faciant. E mediasti-
ni parte illa quæ medios habet pulmo-
nes, profert se membrana egregie spissa,

IN ANATOMICEN

duraq;, qua cor circumquaq; integitur
 περιάπλιον Græcitas nominat. Hæc tue-
 tur cor, ne ab aduenticijs afficiatur, nè ue-
 asperginoso fomento careat, quo feruori
 suo moderetur. Hæc et uiræ cordis unit,
 et halituosos illinc spiritus uehemēti mo-
 tu dissolui prohibet. Hic cor se condit
 princeps membrum, et in turbinem fasti-
 giatū uiscus, tribus intus uentriculis con-
 cauum ac assidue palpitans, cui et suæ
 sunt utrinq; auriculæ in quibus superest
 quam longissime uita. In sinistro cordis
 uentriculo spiritus et exigui sanguinis se-
 des est, à qua uenalis arteria progrediens
 pulmones subit aëremq; ab eis concipit
 præparatiorem, quem in cordis sinus in-
 troducat, ne importunius æstuent. Dex-
 ter uentriculus plurimū et calidissimum
 sanguinem continet. In hunc iecoris uena
 caua per mediam spinam scandit ac uita-
 lis spiritus fomitem infundit. A quo et uer-

duraq̃ , qua cor circumquaꝫ integitur
περικαρδιον Græcitas nominat. Hæc tue
tur cor, ne ab aduenticijs afficiatur, nè ue
asperginoso fomento careat, quo feruori
suo moderetur. Hæc et uireis cordis unit,
et halituosos illinc spiritus uehemēti mo-
tu dissolui prohibet. Hic cor se condit
princeps membrum, et in turbinem fasti-
giatū uiscus, tribus intus uentriculis con-
cauum ac assidue palpitans, cui et suæ
sunt utrinꝫ auriculæ in quibus superest
quam longissime uita. In sinistro cordis
uentriculo spiritus et exigui sanguinis se-
des est, à qua uenalis arteria progrediens
pulmones subit aëremq̃ ab eis concipit
præparatiorem, quem in cordis sinus in-
troducatur, ne importunius æstuent. Dex-
ter uentriculus plurimū et calidissimum
sanguinem continet. In hunc iecoris uena
caua per mediam spinam scandit ac uita-
lis spiritus fomitem infundit. À quo et ue-

INTRODVCTIO.

na arterialis in pulmones copiosum sanguinem eructat. Medio dexteri et sinistri uentriculo, sanguis temperatus, et quantitate mediocris, inest. Ab hoc magna arteria cui Aorte nomen, nascitur uitalis spiritus uehiculum, ea susq; deq; perpetuo agitur contrarijs motibus dilatatione et constrictione, ac secatur demum in ramorum myriadas ut percalereis toto corpore parteis miti flatu refocillet. De mebris uitalibus huc usq; sermonē produximus.

MEMBRANA ossosum capitis orbem forinsecus obducens, περικράσις *perikrasis* Grecis appellatur, et dura est, et spissa et tenax, et exteriori cerebri tunicae, duram matrem eam uulgo uocant, in substantia conformis. Pendet et affixa duræ matris tunica pericranio, sic naturæ uisum est, ne in contactu cerebro efficiat, subter quam et tenuis mollisq; membrana, pia mater ei nomen est, cerebrum in:

na arterialis in pulmones copiosum sanguinem eructat. Medio dexteri et sinistri uentriculo, sanguis temperatus, et quantitate mediocris, inest. Ab hoc magna arteria cui Aorte nomen, nascitur uitalis spiritus uehiculum, ea susq̃ deq̃ perpetuo agitur contrarijs motibus dilatatione et constrictione, ac secatur demum in ramorū myriadas ut percalēteis toto corpore parteis miti flatu refocillet. De mēbris uitalibus huc usq̃ sermonē produximus.

MEMBRANA ossosum capitis orbem forinsecus obducens, περικρά-
viov Grecis appellatur, et dura est, et spissa et tenax, et exteriori cerebri tunicæ, duram matrem eam uulgo uocant, in substantia conformis. Pendet et affixa duræ matris tunica pericranio, sic naturæ uisum est, ne in contactu cerebro efficiat, subter quam et tenuis mollisq̃ membrana, pia mater ei nomen est, cerebrum in-

IN ANATOMICEN

uoluit et nutrit, crebris uenis aspersa,
Dure matri et ipsius cerebri substantiæ
continuatur, et cerebri uentriculos pene-
trat. Hinc se, proxime, oculis offert ipsum
cerebrum, et eius uentriculi, et postico ca-
pitis inditum cerebellum, a quo et me-
dulla spinæ in uertebras descendit. Hinc
et plexus reticularis (rete mirabile triuia-
libus uocatur) summo cerebello, è crebris
uenarum ac arteriarum mutuo sese can-
cellantibus filamentis, phantasiam sui
præbet, in quo spiritus uitalis a corde sur-
sum uectus per arterias dum plenius co-
quitur rarescit, et animalis fit spiritus: sen-
sus et motus causa in uniuerso corpore.
Neruorum enim fons cerebrum est ner-
ui uero spiritus animalis sunt deferentia ua-
sa, qui à medulla spinæ (Nucha barbaris
dicitur) in omnes corporis particulas di-
geruntur. Quin et à cerebro septem neruo-
rum coniugationes procædunt. Bini ner-

uoluit et nutrit, crebris uenis aspersa.
Dure matri et ipsius cerebri substantiæ
continuatur, et cerebri uentriculos pene-
trat. Hinc se, proxime, oculis offert ipsum
cerebrum, et eius uentriculi, et postico ca-
pitis inditum cerebellum, a quo et me-
dulla spinæ in uertebras descendit. Hinc
et plexus reticularis (rete mirabile triuia-
libus uocatur) summo cerebello, è crebris
uenarum ac arteriarum mutuo sese can-
cellantibus filamentis, phantasiam sui
præbet, in quo spiritus uitalis a corde sur-
sum uectus per arterias dum plenius co-
quitur rarescit, et animalis fit spiritus: sen-
sus et motus caussa in uniuerso corpore.
Neruorum enim fons cerebrum est ner-
ui uero spiritus animalis sunt deferētia ua-
sa, qui à medulla spinæ (Nucha barbaris
dicitur) in omneis corporis particulas di-
geruntur. Quin et à cerebro septem neruo-
rum coniugationes procædunt. Bini ner-

INTRODVCTIO.

ui antrorsum nareis spectant, olfactus nimirum organa. Bini ad oculos feruntur, in itinere sese interfecantes, è quibus uidentis facultas. Alij bini motum oculis, bini alij linguae motum et gustum tribuunt. E duobus et uentriculis sensu pollet, quo minus appetentia illi desit, è totidem et exilibus nervis sapes discernit palatum. Vnus postærum nervus utrinque porrectus ab uno principio, auribus largitur dextrae et sinistrae, ne surditate extundantur. Hæc sunt quæ de membris animalibus abs me per compendium dicta, introductione hanc in Anatomien iusta proximitate finiât. Cætera enim quæ ad hanc tractationem pertinent, in alio opere prosequemur: ubi ad Anatomice omneis numeros sermonem accommodabimus.

EXCVDEBAT ROB. REDMAN
nus Londini Anno

M. D. XXXII.

CVM PRIVILEGIO.

ui antrorsum nareis spectant, olfactus ni-
mirum organa. Bini ad oculos feruntur,
in itinere sese intersecantes, è quibus uidē-
di facultas. Alij bini motum oculis, bini a-
lij linguæ motum et gustum tribuunt. E-
duobus et uentriculis sensu pollet, quo mi-
nus appetentia illi desit, è totidem et exili-
bus nervis sapes discernit palatum.
Vnus postræmum nervus utrinq̃ por-
rectus ab uno principio, auribus largitur
dexteræ et sinistræ, ne surditate extundan-
tur. Hæc sunt quæ de membris animalibus
abs me per compendium dicta, intro-
ductionē hanc in Anatomicen iusta pro-
lixitate finiāt. Cætera enim quæ ad hanc
tractationem pertinent, in alio opere pro-
sequemur: ubi ad Anatomices omneis nu-
meros sermonem accommodabimus.

EXCVDEBAT ROB. REDMA-
nus Londini Anno

M. D. XXXII.
CVM PRIVILEGIO.

53

***To the Distinguished and Illustrious
Henry, Earl of Surrey.
David Edwardes, Physician, Sends Greetings***

How often, Henry, I have recalled the honourable achievements of those noble dukes, in what great honour all Englishmen held your grandfather during his lifetime for his remarkable ability and happy successes in warfare, as well as his extraordinary prudence in the administration of civil affairs; and also at present how expertly everything that pertains to us English is daily managed by your famous father. I cannot sufficiently

admire your family, but not so much for those reasons as because I see you established above what can be said for many other young men in this age, and turning your mind so seriously to those things which will render it better. I am by no means certain whether I ought to ascribe this to the benefit of that stock from which you have been brought forth to us, whether to the gods who through you smile upon and favour us English. However it may be, let us hope what has occurred will be to the advantage of our commonwealth, and that the more so since you have pursued worth-while things for so long a time. Thus you will approach the next age better prepared, and good habits will meanwhile strengthen your mind so that later you will not easily fall into worse. But the more you may be strengthened by counsel and prudence, with confidence placed in your family, so much the better guidance will Norfolk have when you succeed as heir to your father's estates. Meanwhile how much more useful you will be to your 54 people as Earl of Surrey, and finally so much the more will all Englishmen desire you to undertake the affairs of the commonwealth. There is no doubt that you can achieve all these things which will be to the increase of your honours and to the honour of your family.

As your talent and gravity of character promise, so we have great hope that you will be like your father and grandfather. I wish you both the greatest successes and the most fruitful increase of all the best things. And once more I wish that this whole year from its beginning may be happy for you and yours. With this augury I dedicate to you this our introduction to anatomy. For as this part of the art of medicine is not known to all, because it is something very difficult to comprehend, it requires an easy arrangement by which readers, as if led by the hand to it, may lean upon it. This is indeed a slight work, but wholly useful for all physicians and surgeons, because it explains many things briefly. It contains nothing obscure, nothing elaborate, very readily accessible to the talents of all those who are neither dull nor ill-adapted to matters of knowledge. In this, if anything differs from the common opinions of physicians, let no one be astonished because the learned do not believe the same in these matters.

Hereafter, if God permit, I shall compose a complete book of anatomy in which I shall further the opinions of all the learned, to which my own opinion will be added. I could have done this at present but not, however,

with the same effort or with the form of an introduction preserved. It remains that this little book, which we have enlisted in the service of the commonwealth, may be pleasing to you, for it recognizes the existence of those very few unlearned physicians by whose mistakes many perish, from which this fact will be gathered, that no parts of the body should be unknown to physicians. Farewell. Cambridge. 1 January.

THE INTRODUCTION TO ANATOMY OF DAVID EDWARDES ENGLISHMAN

The whole lower venter—for thence it is necessary to begin the dissection of the human body because that part putrefies very readily—from the outer skin to the peritoneum is called EPIGASTRION by the Greeks and *mirach*^[1] by the Barbarians, of which the following are the parts.

The superficial skin which covers the whole body is completely insensitive. The skin lying and stretched under the very thin and superficial skin is sensitive.^[2] The Greeks call it HYPODERMA. A kind of fat occupies the whole venter and is spread under the sensitive skin except for the middle.

A sinewy and thin membrane immediately follows this. A membrane taking origin from the muscles is subjoined firmly to this where a straight line appears in the middle.^[3]

Two oblique descending muscles^[4] lie under these toward the lowest venter.

The oblique ascending muscles^[5] are placed under these. Two rectus muscles have a close relationship. And lowest of all are the transverse^[6] 56

muscles. Therefore there are these eight for which there are nearly individual sinewy coverings by which they are distinguished from one another.

An aponeurosis, or rather a thick and firm membrane, is stretched under them which some falsely call the peritoneum. So much for the epigastrium and its parts.

Certainly the peritoneum is a sinewy part, soft to the touch, of ordinary firmness, occupying the whole venter, and resting under the aponeurosis or membrane which I mentioned. The Greeks gave that name to it. The Barbarians call it *siphac*.

The *zirbus* or omentum is extended under the peritoneum. The *zirbus* is a kind of fat derived from sinewy threads and the slender adipose substance of the nerves; it is less thick than the fat previously mentioned. It covers much of the intestines and the lowest part of the stomach and assists the coction of aliment.

The intestines take origin from the stomach; of them, that which is called rectum and *longanon* ^[7] is the lowest of all the intestines and contains the dry burden of the bowel, and its head extends outward between the nates so that it may dispose of its burden. The colon is continuous with it and in its ascent goes around the left kidney, and at the sides of the stomach it falls away to the right. ^[8] What the Greeks call TYPHLON and MONOPHTHALMON, the Romans the blind intestine and one-eyed, ^[9] is attached to the colon, of which it is the only passage; for the other end is closed so that it may assist coction more suitably in the manner of the stomach. Hence the name for the thing. And such is the number of the thicker intestines. 57

The caecum is continuous with the ileon, ^[10] an intestine twisted into numerous sinuses; from its shape the Greeks gave it the name PARA TOU EILEISTHAI, that is, from its involvement; and its disease is called *iliacus*. The jejunum follows it. Dissectors of bodies gave this name jejunum to the latter intestine because of the fact that it is always found empty and contains nothing. For the liver first snatches away whatever the jejunum might contain. Above all these intestines arises the duodenum which is continuous below with the jejunum and above with the pylorus. It is called DŌDEKA DAKTYLOM by the Greeks from the measure of twelve fingers. These three [intestines] by reason of their substance are called the slender intestines.

The stomach is located under the diaphragm, of which the upper mouth ends in the oesophagus, properly called *stomachus*; the lower opening

through which aliment is sent into the intestines is called PYLÖROS.

The spleen is an organ of rare substance and lies at the left side of the stomach; the liver being in the right hypochondrium. The latter is rounded and to some degree lunate, the former longish and somewhat quadrate. The

gibbous part ^[11] of each of these extends toward the lower ribs, because there is a concavity in each of them which is very close to the stomach. The

liver gives rise to the blood. ^[12] The spleen purges it of black bile. The spleen increases with loss to the rest of the body. The size of the liver is useful to the whole bodily structure, because it provides copious blood 58 and natural spirit. The liver has lobes which the Greeks call LOBOUS,

sometimes three, sometimes more, ^[13] and in its hollow extends the gall bladder by which the blood is freed of bile and issues forth pure. It is especially by exhalation and transpiration of this bladder that the duodenum

and jejunum are sometimes stained; ^[14] sometimes they are irritated if there is a very large transpiration of particularly corrosive bile.

From the hollow of the liver ^[15] arises the portal vein which is formed from the concurrence of the many slender veins of the liver. On the other hand, it divides again into innumerable parts and gives off an immense multitude of veins which afterward are inserted here and there into almost all the intestines and to the little adipose membranes mixed together, so that they provide nutritional substance for the liver in the generation of blood. For chyle and food are sent down from the stomach directly to the intestines; the pylorus yields an exit as soon as the stomach has received as much as suffices for its uses and has accomplished its coction. Unless it be transmuted into the nature of blood [this food] contributes very little toward the nourishment of the rest of the body. Therefore these numerous venules serve to draw out from the intestines the best juice of the nutriment as yet not sufficiently concocted, and deliver it to the hollow of the liver where the blood is made. Doubtless those venules can be called meseraics, or by the

Greek word mesenterics. The Latins call them milk veins. ^[16] For their protection, lest in their numerous ramifications some of them be torn apart or rent by a more vigorous motion of the body, the PANKREAS, that is,

glandular flesh which is sometimes called KALLIKREAS by the Greeks, 59 attaches to the duodenum so that the venules may individually be more firmly supported.

The blood passes from the hollow of the liver, in which it was formed a little earlier, to the gibbosity ^[17] of the liver; however, it is not the same kind as was made in the hollow but more pure and simple, since both biles have been strained from it and transmitted to their receptacles so that the blood may be more unsullied for nourishing the body wholesomely and for producing spirits. From the gibbosity the blood is extended throughout the whole body through the vena cava—called KOILĒ by the Greeks—and by the many branches of that vein. This vein surpasses all the rest of the veins of the body in size and arises from the gibbosity of the liver. Descending from this through the middle of the spine, one [branch] on each side seeks the kidneys, each branch extending a palm's length.

These branches of the vena cava are the emulgent veins. ^[18] In the body of that one whom we dissected very recently the left branch had a higher place of origin. ^[19] Very often, however, the opposite occurs, so that the right emulgent vein is carried higher in the body. Nature employs these emulgent veins for carrying down the watery part and bile of the blood from the liver to the kidneys. A like number of little branches of arteries in the same site, from the great aorta artery going under the vena cava, run an equal length into the kidneys under the emulgent veins, unburdening the heart of 60 bile and watery blood; these have the name of emulgent arteries.

A vein and artery descend from the left emulgents into the testis of the left side. They are the seminal passages swollen with blood and spirit; the seminal matter contained in them procreates females, because their humour is watery and requires coction. Seminal passages, likewise an artery and vein, are extended downward from the right [emulgents] into the right testis; but having arisen from the trunks of the vena cava and aorta artery, therefore the juice in them is less watery, and properly concocted is more suited for the generation of males. In these passages blood is concocted, and

afterward transferred to the glandular flesh of the testes it acquires the form of semen. ^[20]

The kidneys are solid and hard organs, not sentient, and the attractive force in them is very powerful. They purge the blood of its watery part and bile, but they retain [some of] the blood so that they may be nourished by it and expel the rest of the humour. For the OURÊTÈRES are attached to them, ^[21] that is, the urinary passages, whitish, reed-like and tensile which it may be said extend to the bladder and are similar to its substance.

The diaphragm is a membranous substance, running between the vital and natural members. It is called DIAPHRAGMA by the Greeks. It strengthens the expulsive force in the intestines, it is assigned to the members selected for spirit, and it curbs the smoky vapours lest they blacken the vigorous spirits of the heart and brain. Above, there is affixed to it a sinewy covering which clothes the thorax inwardly and binds the pectoral ribs to the interstitial spaces, which covering the Greeks in good part name ^[61]

PLEURA, but sometimes it is called HYPOZŌMA ^[23] by them. By its inflammation pleurisy occurs, the name taken from the covering.

From the pleura near the spine arises a membrane separating the lungs and lower thorax into equal parts through the middle. It is commonly called the mediastinum, and is so well adapted to the lungs that a defect of one lung is not easily communicated to the other. ^[24]

Certainly the lungs inhabit the middle palace of the chest, invigorate the spirits of the heart and brain, temper the heat and avert the danger of suffocation, and have lobes like the liver. They hold the heart constantly in a kind of embrace in the manner of very caressing nurses and sing a harmony of qualities by which they soothe the individual parts of the body and make them vigorous. From that part of the mediastinum which holds the middle of the lungs, a thick and hard

^[25] membrane appears which completely covers the heart, called in Greek PERIKARDION. This protects the heart lest it be afflicted by accidental things; and lest it lack the moistening fomentation by which its heat is moderated. It unites the forces of the heart and prevents the exhaled spirits from being

dispersed by vehement motion. Here the heart establishes itself, prince of members ^[26] and an organ sharpened into [the shape of] a top; hollow within; continuously palpitating by its three ventricles, ^[27] with an auricle on each side in which life remains the longest. ^[28] The seat of the spirit 62 and a small amount of blood is in the left ventricle of the heart, from which the pulmonary vein advances and enters the lungs to receive better-prepared air from them; ^[29] this it introduces into the ventricles of the heart lest they become unduly heated. The right ventricle contains more and very hot blood. The vena cava rises into this ^[30] through the middle of the spine and pours in the tinder of the vital spirit from the liver. From this the pulmonary artery belches much blood into the lungs. In the ventricle between the right and left there is tempered blood of slight quantity. From this ventricle the large artery called the aorta arises, the vehicle of the vital spirits; it is constantly agitated up and down by the contrary motions of dilatation and constriction, and finally it is divided into myriads of branches so that it revivifies the living parts in the whole body by a gentle flatus. This is the end of the account of the vital members.

The membrane covering the bony roundness of the head outwardly is called PERIKRANION by the Greeks, and it is hard, thick and firm, and conforms in substance to the exterior covering of the brain which is commonly called the dura mater. The covering of the dura mater hangs affixed to the pericranium, so it seemed to nature, lest in contact it have an effect on the brain; under this covering a thin and soft membrane, which is called 63 the pia mater, sprinkled with numerous veins, envelops and nourishes the brain. It is continuous to the dura mater and the substance of the brain, and it penetrates the ventricles of the brain. Hence the brain displays itself very clearly to the eyes, both its ventricles and the cerebellum placed at the rear of the head from which the medulla descends into the vertebrae of the spine. Here the reticular plexus (commonly called the *rete mirabile*), woven together from numerous slender threads of veins and arteries at the summit of the cerebellum, displays its *phantasia*; in it the vital spirit carried upward from the heart through the arteries, having been fully concocted and rarefied, becomes animal spirit, the cause of sensation and motion in the

whole body. For the brain is the source of the nerves, but the nerves are the vessels which distribute animal spirit; ^[31] from the medulla of the spine (it is called *nucha* by the Barbarians) they are distributed to all parts of the body. Furthermore, there extend from the brain seven pairs of nerves. ^[32] Two nerves look forward to the nares, ^[33] the olfactory organs. Two are carried to the eyes, ^[34] intersecting in their course, from whence comes the 64 faculty of vision. Another two [carry] motion to the eyes, ^[35] another two give motion and taste to the tongue. ^[36] From two the stomach acquires sensation ^[37] so that appetite may not be lacking to it, and from as many slender nerves the palate distinguishes flavours. ^[38] Finally, from a single origin one nerve is extended on each side, provided for the right and for the ^[39] left ear lest they be struck by deafness. These things which have been said by me briefly regarding the animal members, within the proposed limits, end this introduction to anatomy. Other matters which pertain to this subject I shall discuss in another work where we shall adapt the discourse to all aspects of anatomy.

Printed by Rob. Redman in London
M.D.XXXII
With Privilege

Footnotes

[1] The term *mirach* means the anterior abdominal wall, but here Edwardes refers to the abdominal wall as venter. Lower venter proper means the abdomen or abdominal cavity together with the pelvis.

[2] There is confusion here for, of course, the skin of the body is most sensitive. The subcutaneous tissues, on the other hand, are relatively insensitive.

[3] He refers to the aponeurosis of the external oblique muscle. It joins with its fellow of the opposite side in the mid-line at the linea alba.

[4] External oblique muscles.

[5] Internal oblique muscles.

[6] Transversus abdominis.

[7] *Longanon* is the medieval Latin term for rectum. In the text which follows it will be noted that Edwardes describes the intestines from below upwards.

[8] The hepatic flexure of the colon.

[9] The caecum was termed the *monoculus* by medieval anatomists. There is no mention of the appendix; this was first described in 1523 by Berengario da Carpi.

[10] The ileum.

[11] The curved surface.

[12] Galen maintained that the major veins had their origin in the liver.

[13] Multiple lobes to the liver was another teaching of Galen derived from comparative anatomy.

[14] Post-mortem staining of the viscera with bile is very common.

[15] The porta hepatis.

[16] This is a good account of the function of the lymphatic vessels.

[17] The gibbosity of the liver is its curved, upper surface.

[18] The emulgent veins are the renal veins.

[19] This is normal in man but in some animals the right renal vessels arise higher than the left. It will be noted that he speaks from his own experience. It is a pity that he qualifies this statement in the next

sentence with a reference to the then current teaching, derived from Galen.

[20] The fallacious idea of the testis filtering off the sperm from the blood brought down to it by the testicular artery lasted a long time. Note too the old fallacy of the left testis producing a female foetus and the right producing a male.

[21] The ureters.

[22] The pleura.

[23] Either Edwardes or the printer was at fault in the form of the Greek script, while the word has more the meaning of diaphragm than of pleura.

[24] Edwardes is obviously aware of the individuality of each pleural sac.

[25] The pericardium.

[26] Note that the heart is the most important organ of the body.

[27] The three-ventricled heart was a myth which remained entrenched in anatomy until Niccolò Massa (1536) and Vesalius. Leonardo da Vinci showed that there were only two ventricles but his drawings were not seen by his contemporaries.

[28] Edwardes is to a degree correct when he says that life remains longest in the auricles. Slow contraction of the auricles can be seen for a short

time after contraction of the ventricles has ceased. This passage could suggest that he practised vivisection.

[29]

Note the persistence of the old idea that the left ventricle contains air.

[30]

The medieval anatomists regarded the right and left atria as part of the corresponding ventricle, hence they stated that the venae cavae opened into the right ventricle. Edwardes's acceptance of the old theory is interesting for just above this in the text he mentions the auricles as separate chambers.

[31]

Galen taught that the nerves were hollow and carried the animal spirit from the brain to the periphery. The vital spirit (air) was carried by the arteries to the brain where, in the *rete mirabile* it was transformed into the animal spirit.

[32]

The ancient idea that there were seven pairs of nerves did not disappear from anatomical teaching until Thomas Willis in 1664 increased the number to nine and Samuel Thomas Soemmerring in 1778 established the modern order of numbering the nerves into twelve pairs. In the account which follows Edwardes does not follow the ancient description of the cranial nerves. According to Galen, and indeed Vesalius, the olfactory nerves were not regarded as separate entities; moreover the glossopharyngeal, vagus, and accessory nerves were part of a single nerve. Edwardes does not describe the trigeminal or facial nerves nor the trochlear or abducent. The trochlear nerve had been described by Alessandro Achillini in 1520. The abducent nerve was to be described later by Eustachius.

[33]

Olfactory nerves. That Edwardes regarded them as functional units is worthy of note.

[34] Optic nerves (the ancient first pair).

[35] Oculomotor nerves (the ancient second pair).

[36] A combination of the modern hypoglossal and trigeminal nerves (the seventh and third pairs).

[37] Vagus nerves (part of the sixth pair).

[38] Glossopharyngeal nerves (part of the sixth pair).

[39] Auditory nerves (part of the fifth pair).

Transcriber's Notes

- Retained publication information from the printed edition: this eBook is public-domain in the country of publication.
- Corrected a few palpable typos.
- In the text versions only, text in italics is delimited by _underscores_.

*** END OF THE PROJECT GUTENBERG EBOOK INTRODUCTION
TO ANATOMY, 1532 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.